Prepared by: Margaret Burkes
Presented by: William Mancuso

Solar-Powered Rover

By Team 66:

William Mancuso (EE), Bowen Williamson (EE), Margaret Burkes (EE), Keldon Ngo (EE), Kenny Bui (EE)

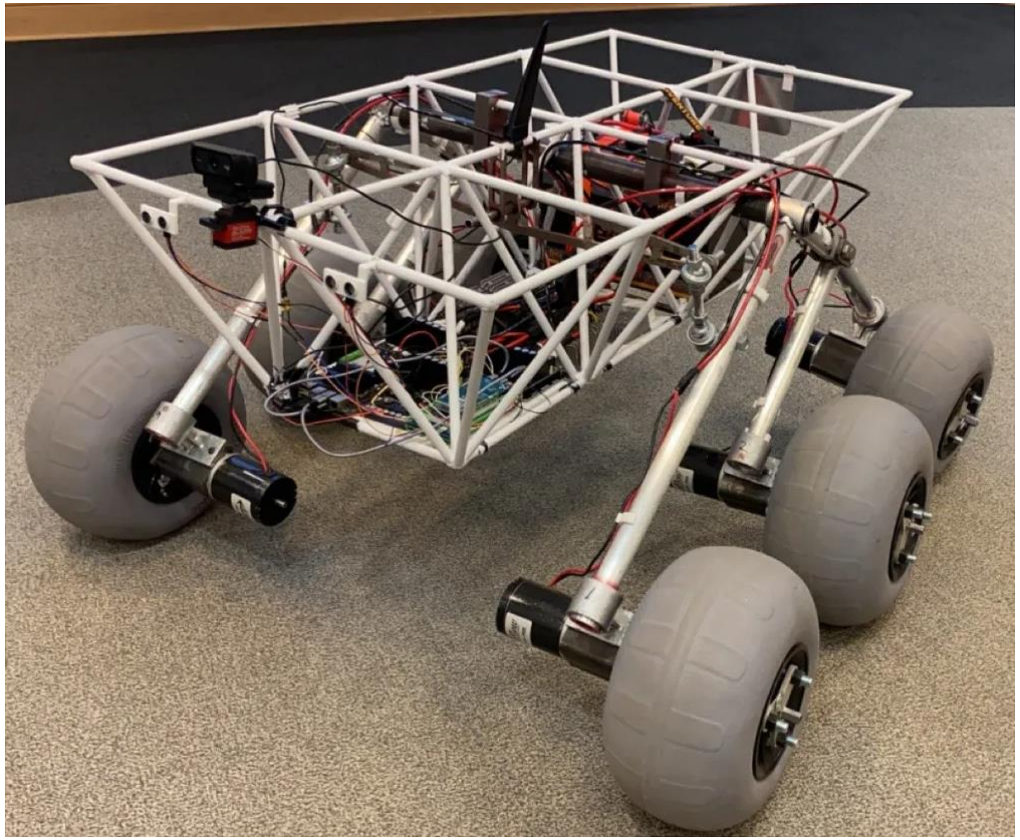
Project Sponsor: Dr. Xiangyu Meng

Faculty Advisor: Dr. Shahab Mehraeen

Prepared by: Margaret Burkes
Presented by: Margaret Burkes

Project Background:

Land
Optimized
Upgradeable
Interplanetary
System



Prepared by: Margaret Burkes
Presented by: Margaret Burkes

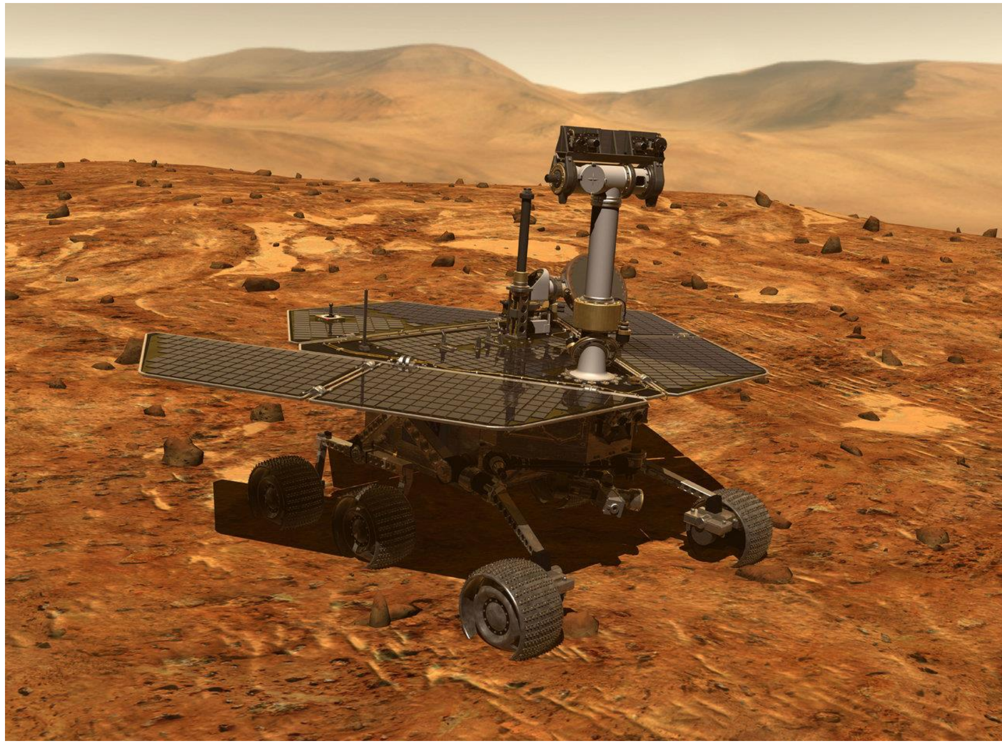
Objective Statement

The main objective of this project is enhancing the previously built rover, which is battery powered, by integrating solar power, autonomous power switching between solar and battery power, and an independent cleaning system for the solar panel.

Prepared by: Keldon Ngo
Presented by: Keldon Ngo

Existing and Competing Technologies:

- Nasa Spirt Rover:
  - Key Similarities
    - Mainly solar power rover
    - Used for environmental exploration
  - Key differences
    - Built for Martian weather
    - No Self-Cleaning System



Prepared by: William Mancuso
Presented by: Kenny Bui

Functional Requirements

#	Weight	Function	Explanation
F1	0.4	Convert solar energy to power the rover.	Utilize solar panels to absorb sunlight as an additional power source to the battery power.
F2	0.35	Autonomously switch power between power sources.	Autonomously switch between battery and solar power.
F3	0.25	Clean the solar panels.	Intermittently clean off debris from solar panel to maximize energy absorption and efficiency.

Prepared by: William Mancuso
Presented by: Kenny Bui

Qualitative Constraints

#	Weight	Qualitative Constraint	Explanation
Q1	0.40	Compatibility with Existing Design	Ensure the implementations to the previous rover are compliant with each other. The additions should merge with the existing design seamlessly.
Q2	0.35	Weatherproof Design	The solar panel system should be built to withstand the elements of Louisiana such as wind, precipitation, and heat.
Q3	0.25	Portability	Make sure that the system doesn’t hinder the transportation of the rover. Should be convenient to maneuver without disassembly.

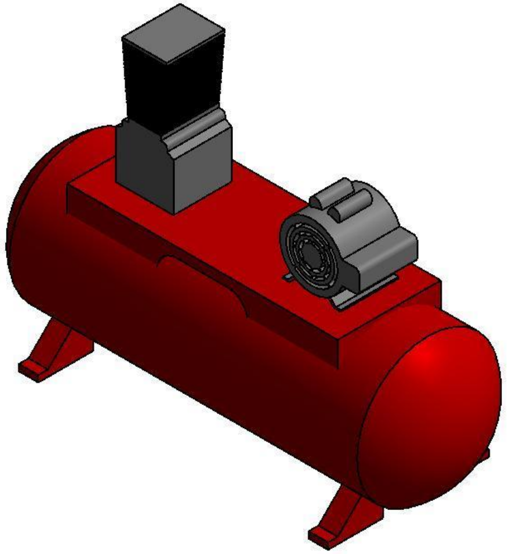
Prepared by: William Mancuso
Presented by: Kenny Bui

Quantitative Constraints

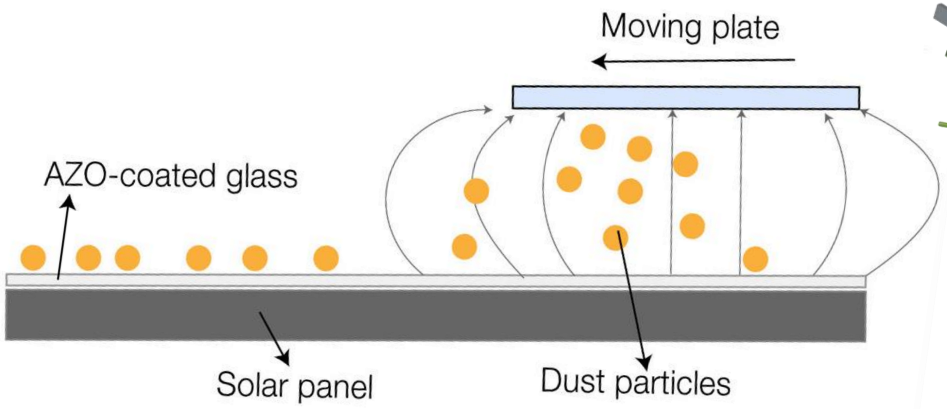
#	Weight	Name	Symbol	Units	Expected Value(s)	Measured Value(s)	Explanation
M1	0.5	Budget	B	$	4000	3525	A budget of $4000 was decided upon by the sponsor.
M2	0.4	Interior Space	S	M^3	0.977	0.971	This was the amount of available space inside the rover's chassis to install new electronic components.
M3	0.1	Weight Limit	W	Kg	<36.3	36.2	The max additional weight load was measured at 36.3kg. The team would like to stay under this variable.

Prepared by: Margaret Burkes
Presented by: Margaret Burkes

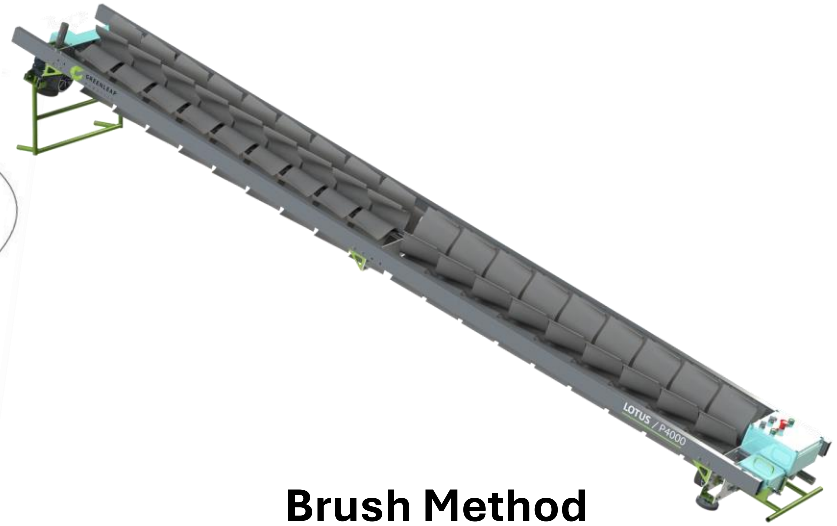
Concept Generation: Self-Cleaning



Air Compressor Method



Electrostatic Method



Brush Method

Prepared by: Margaret Burkes
Presented by: Margaret Burkes

Concept Evaluation & Selection: Self-Cleaning

		Concepts		
Criterion	Weights	Air Compressor	Electrostatic Method	Brush Method
Cost	30	0	-1	1
Cleans Effectively	25	-1	1	0
Mass	20	-1	0	1
Non-abrasive	20	0	1	-1
Maintenance	5	-1	1	0
Weighted Total	100	-0.5	0.2	0.3

Prepared by: Keldon Ngo
Presented by: Keldon Ngo

Concept Evaluation & Selection: Solar Panel

		Concepts		
Criterion	Weights	Thin Film	Polycrystalline	Monocrystalline
Efficiency	40	-1	0	1
Cost	25	1	0	-1
Weight	20	1	-1	-1
Life-span	15	-1	1	1
Weighted Total	100	-.1	-.05	0.1

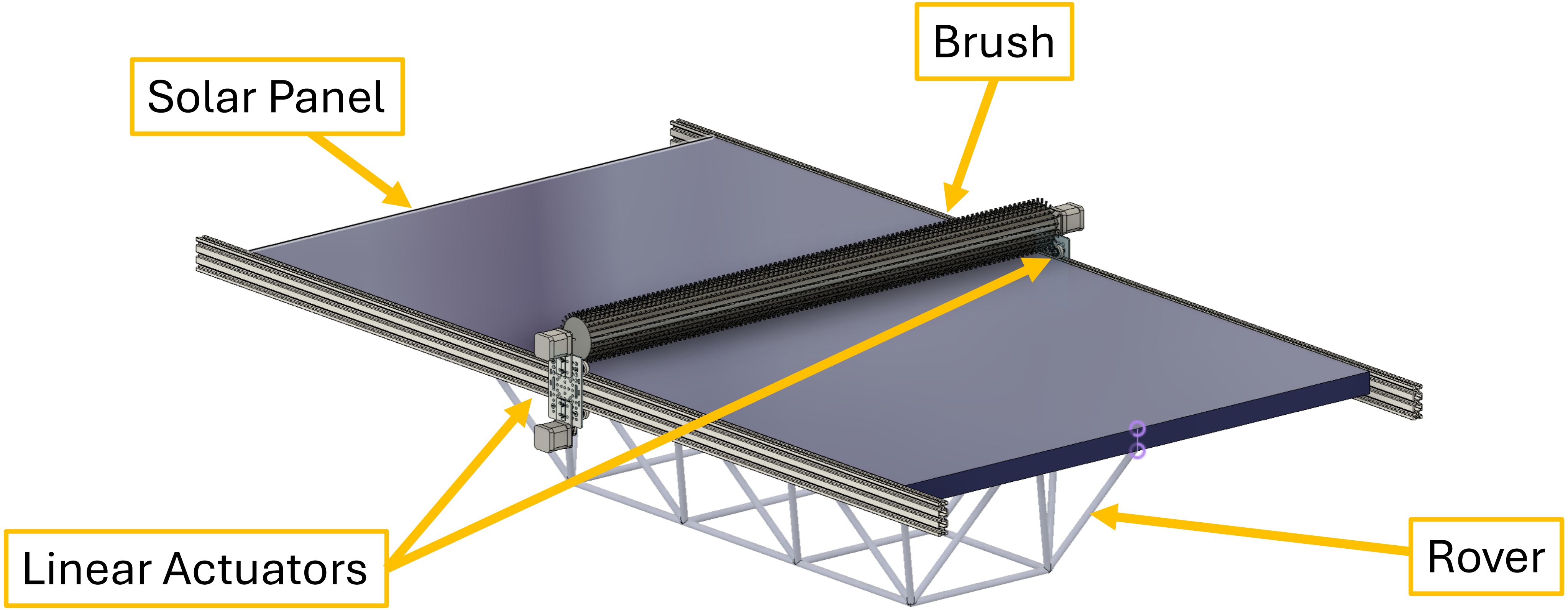
Prepared by: Keldon Ngo
Presented by: Keldon Ngo

Concept Selection:

	Solar Panel Cleaning System	Autonomous Power Flow	Solar Panel
Concepts	Industrial Brush Corp. Brush Method	MorningStar Solar Charge Controller	REC Alpha Monocrystalline panel

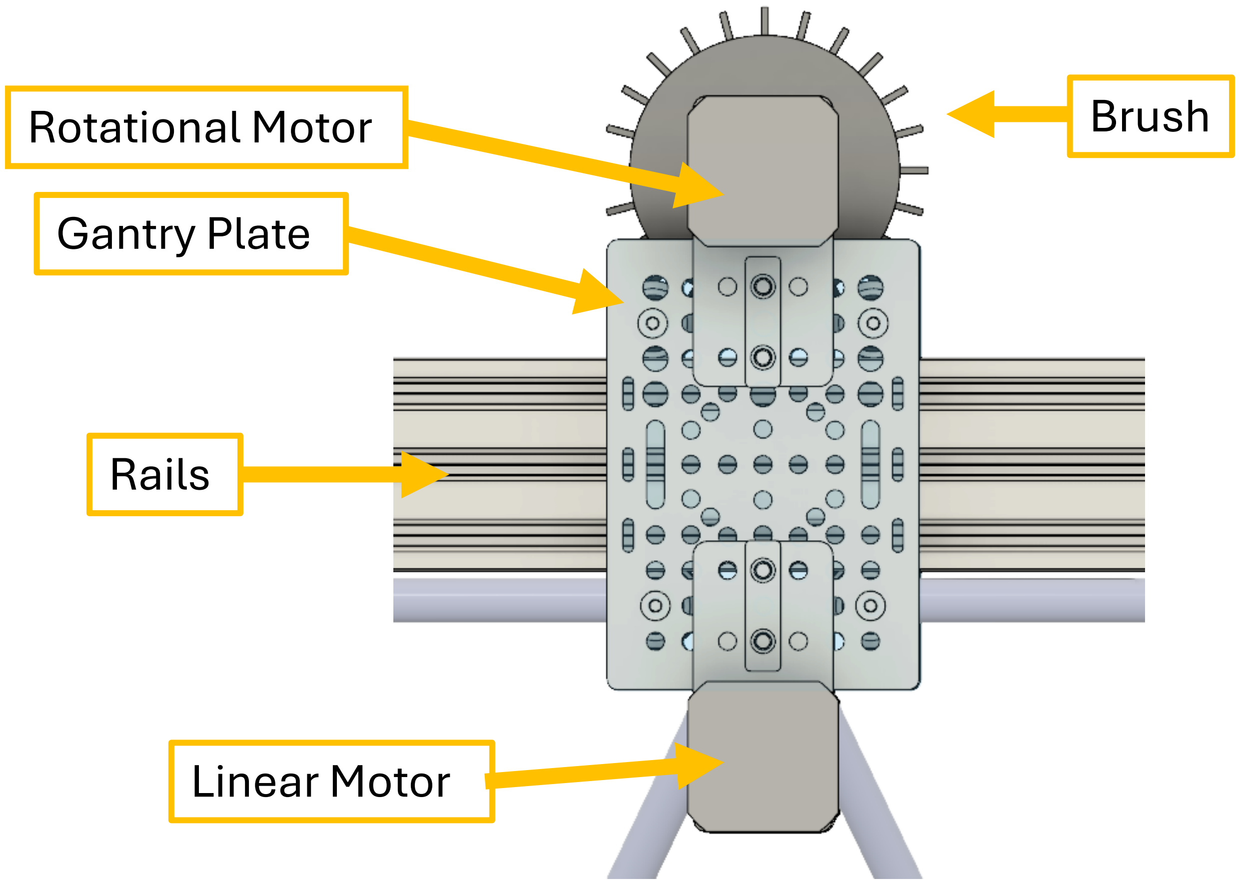
Prepared by: Bowen Williamson
Presented by: Bowen Williamson

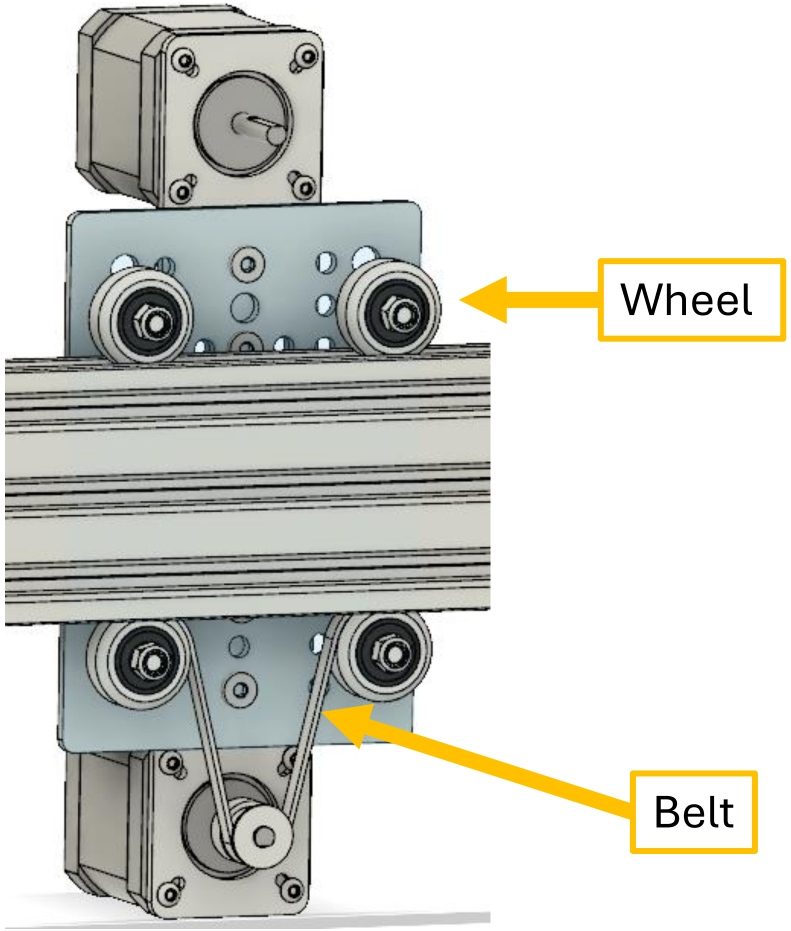
Product Architecture



Prepared by: Bowen Williamson
Presented by: Bowen Williamson

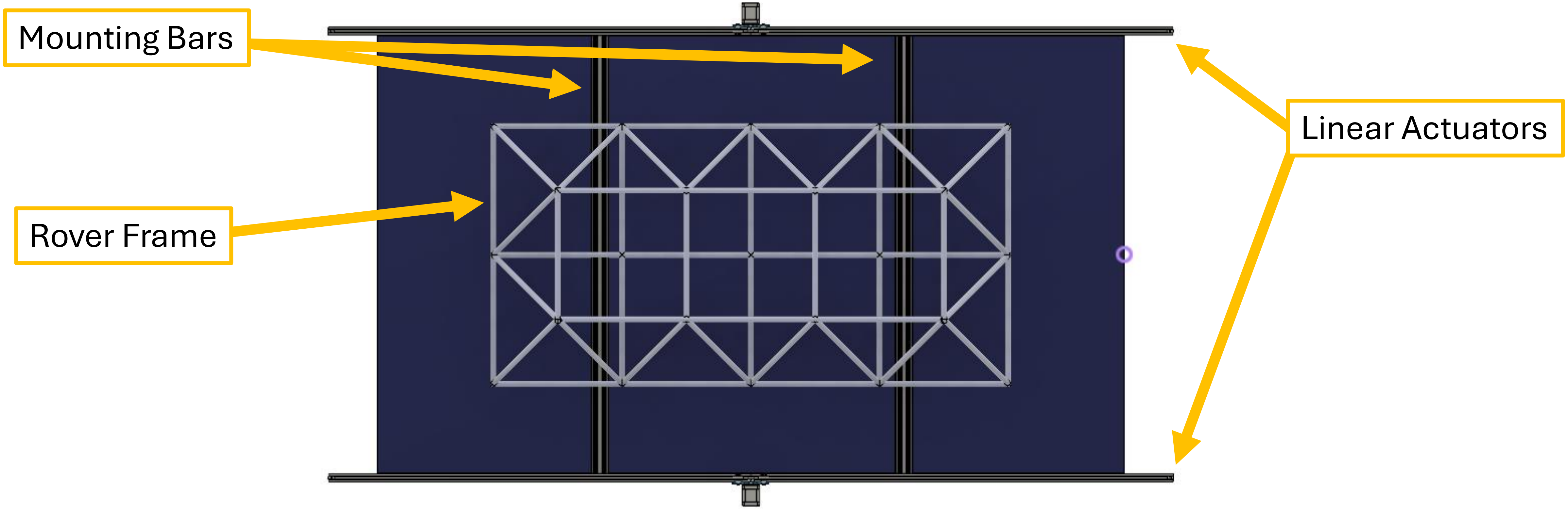
Product Architecture – Linear Actuator





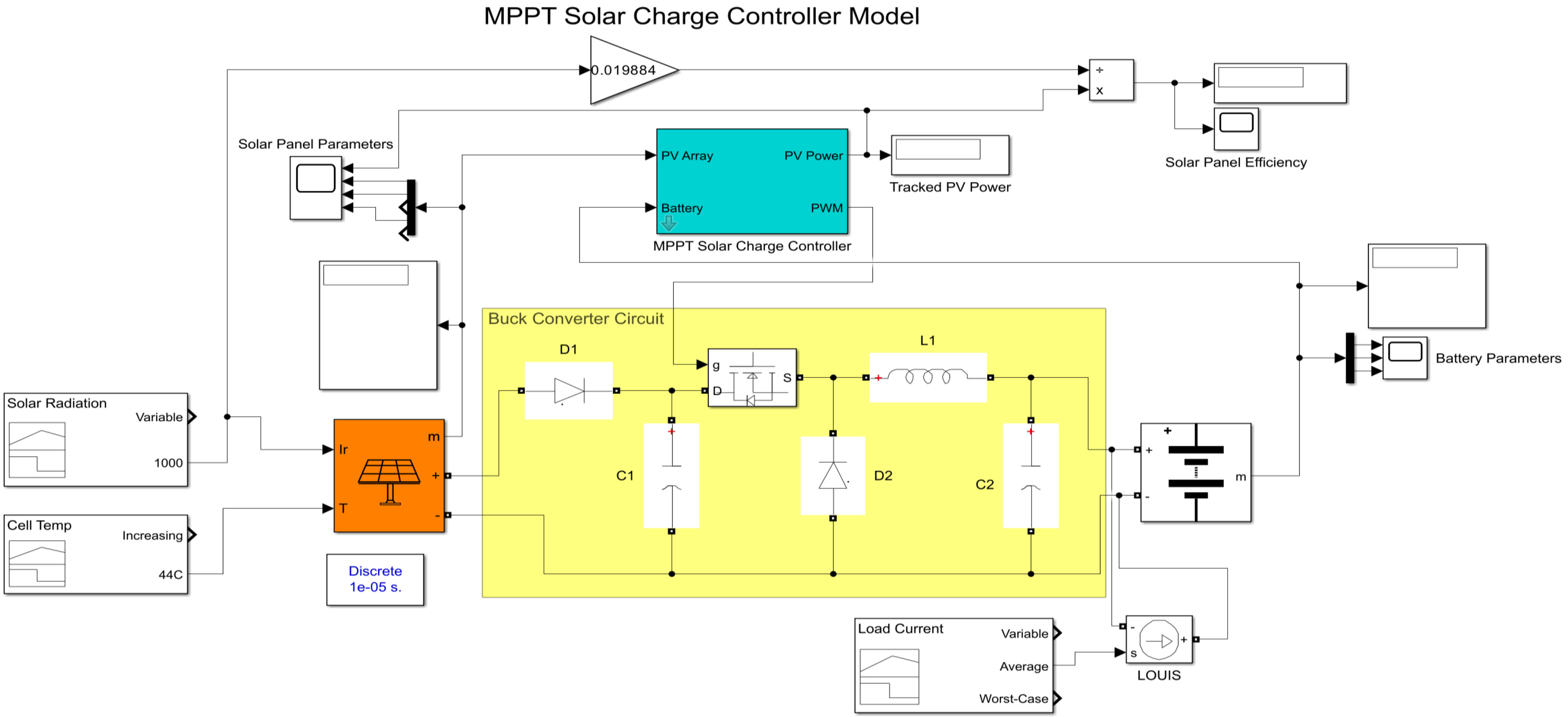
Prepared by: William Mancuso
Presented by: William Mancuso

Product Architecture – Rover



Engineering Analysis: Charge Controller

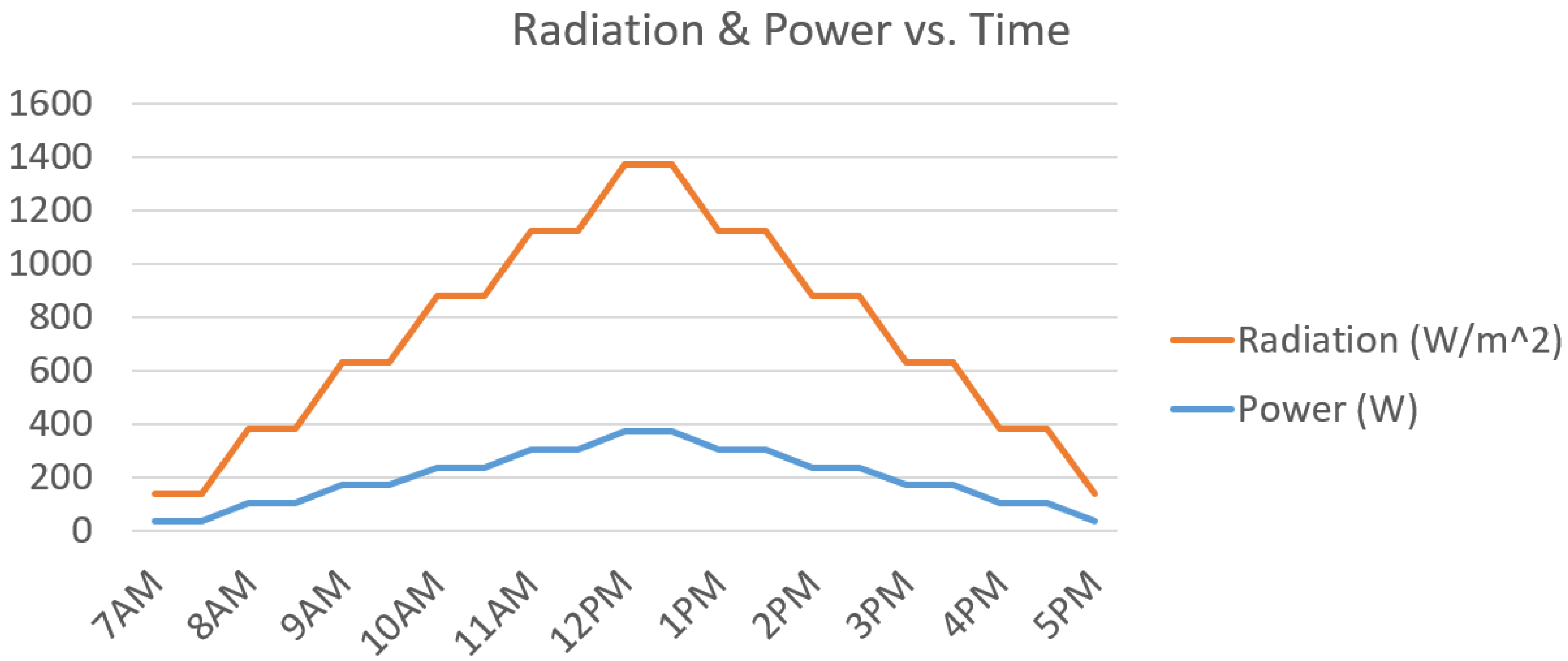
Prepared by: William Mancuso
Presented by: William Mancuso



Prepared by: Bowen Williamson
Presented by: Bowen Williamson

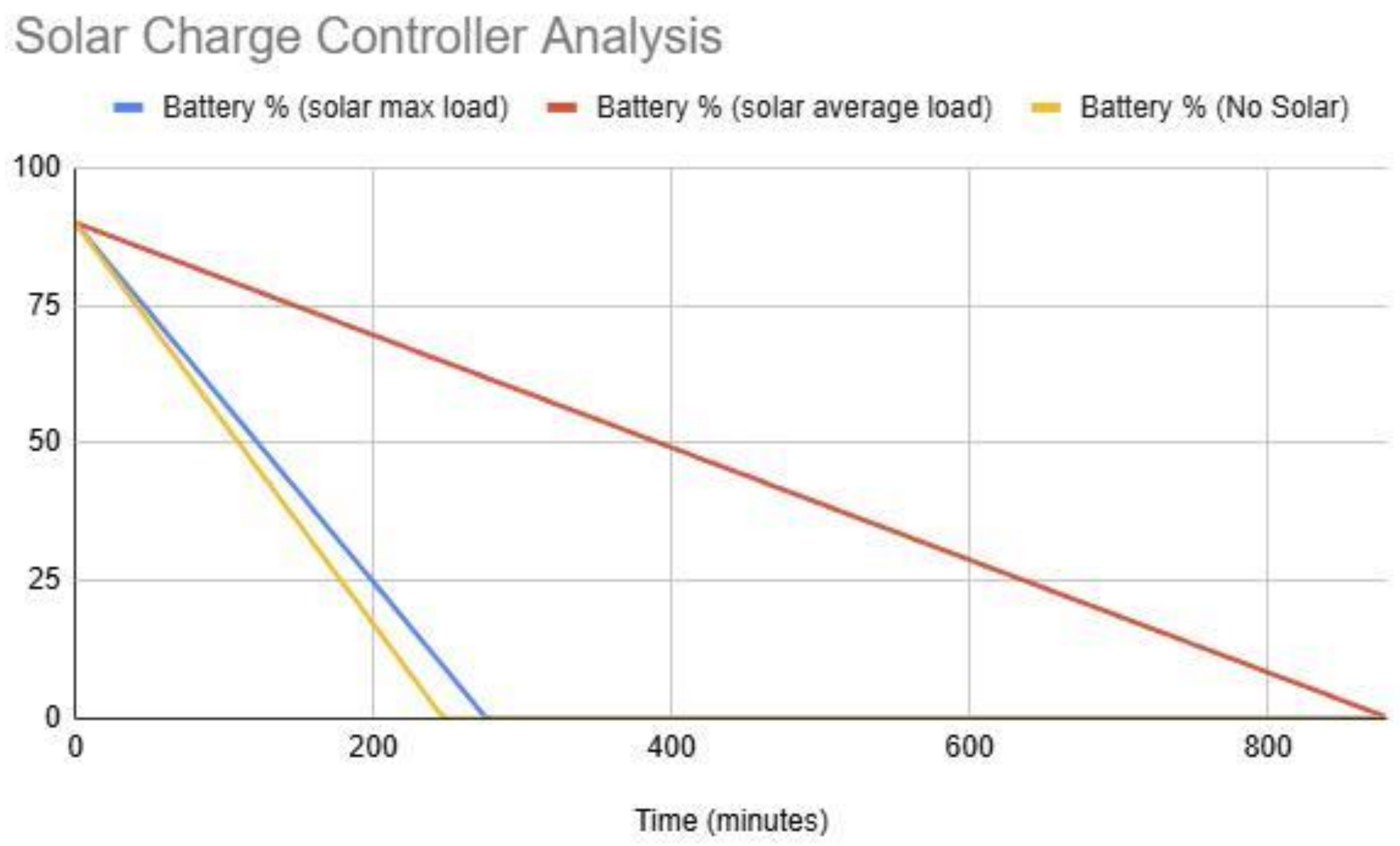
Engineering Analysis: Charge Controller

Effects of Solar Radiation on Power Production



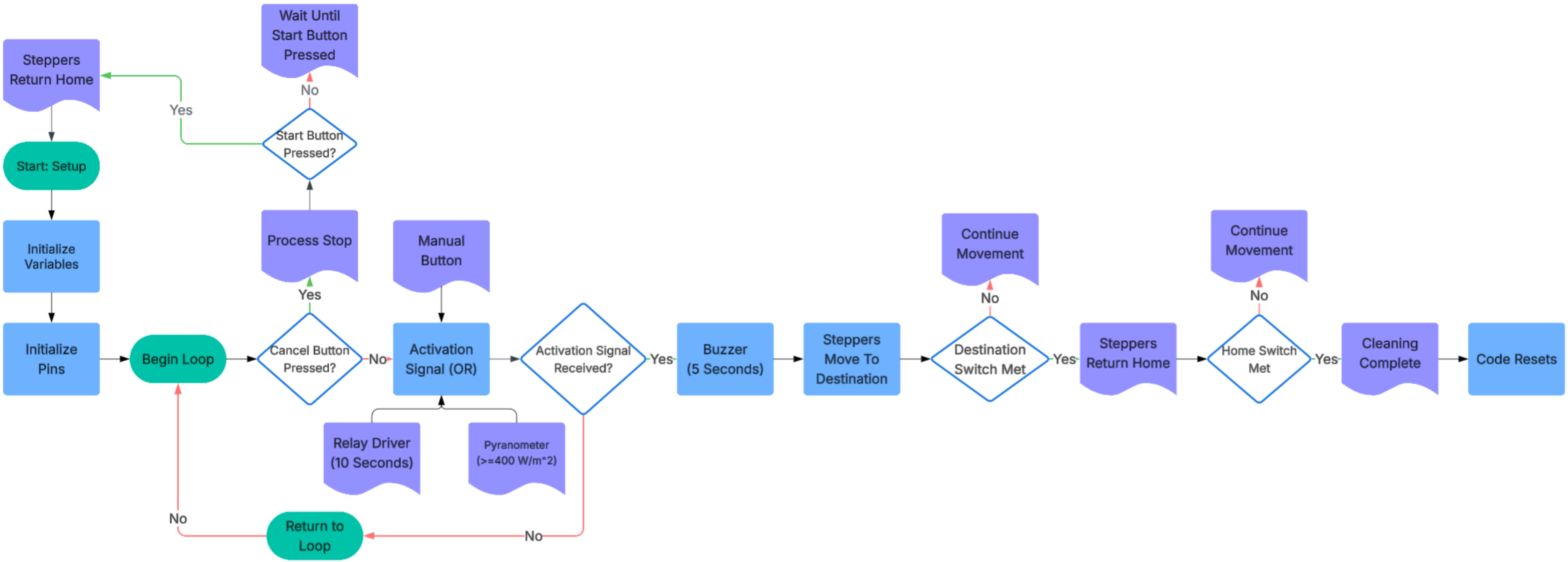
Prepared by: Bowen Williamson
Presented by: Bowen Williamson

Engineering Analysis: Charge Controller



Prepared by: Kenny Bui
Presented by: Kenny Bui

Engineering Analysis: Cleaning Activation



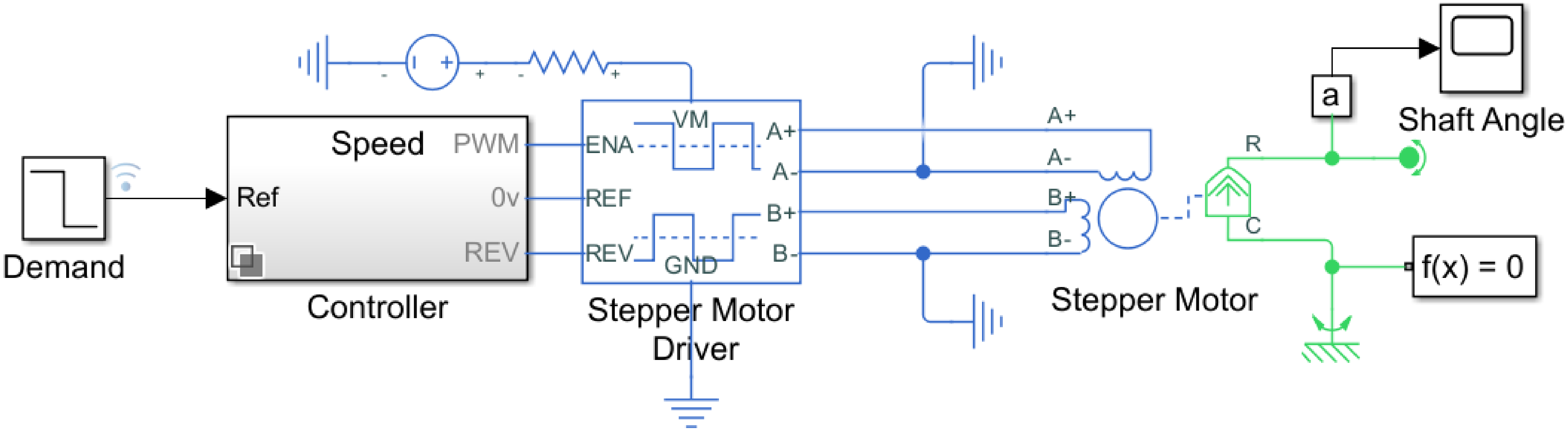
Prepared by: Margaret Burkes
Presented by: Margaret Burkes

Engineering Analysis: Linear Motion Motor

- $M = 3.35kg$
- $\frac{d\omega_m}{dt} = 15.71\, rad/s^2$
- $r = \frac{(\#\, of\ teeth)(pitch)}{2\pi} = \frac{20 \times 3mm}{2\pi} = 9.5mm$
- $T_{em} = \frac{1}{2} M r^2 \frac{d\omega_m}{dt} + r^2 M \frac{d\omega_m}{dt} + \mathrm{r} f_l = 0.31\ Nm$

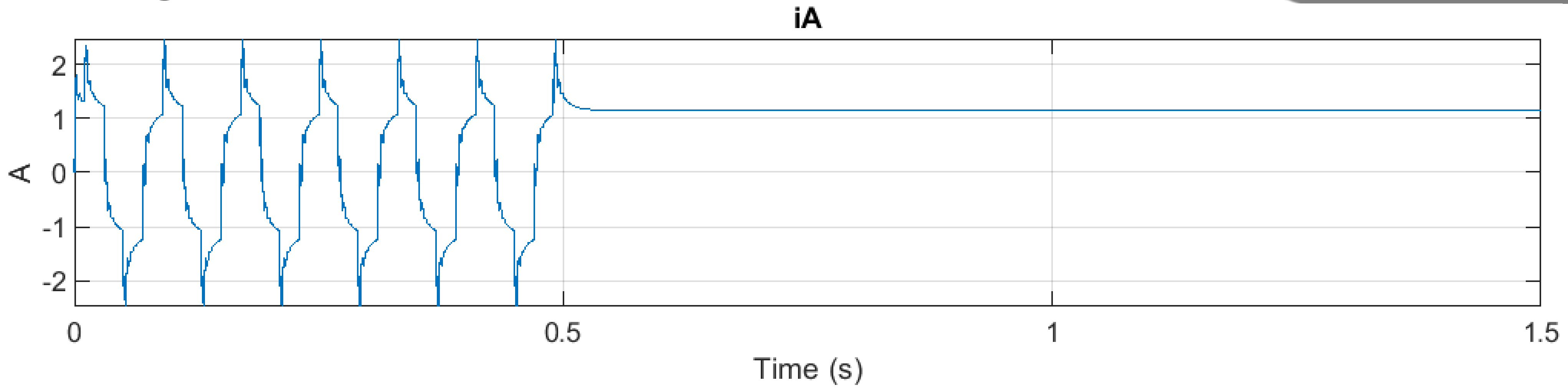
Prepared by: Margaret Burkes
Presented by: Margaret Burkes

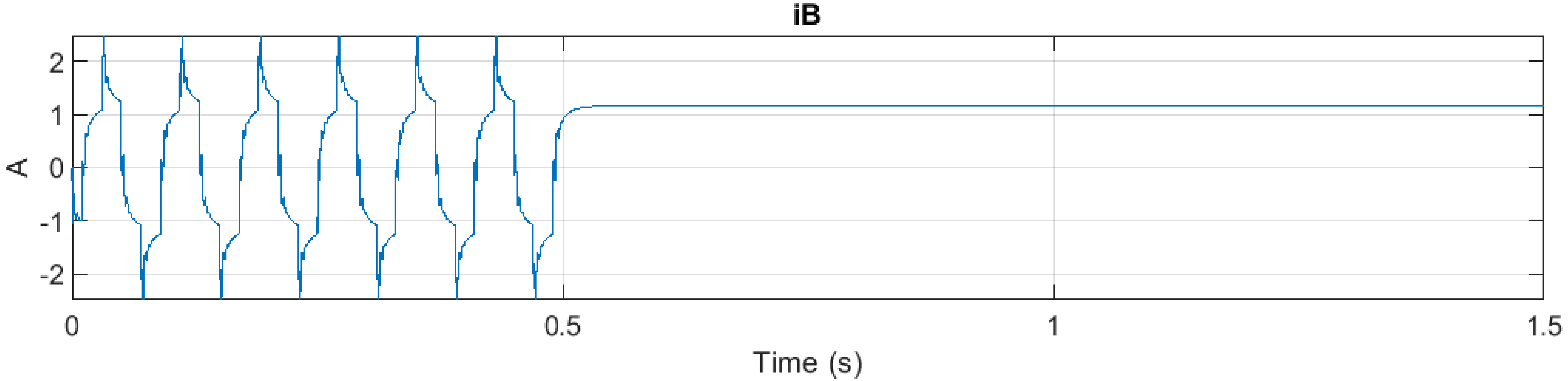
Engineering Analysis: Linear Motion Motor



Prepared by: Margaret Burkes
Presented by: Margaret Burkes

Engineering Analysis: Linear Motion Motor





Prepared by: Keldon Ngo
Presented by: Keldon Ngo

Engineering Analysis: DC Geared Motor

Moment of Inertia

$$I = \frac{1}{3}mL^2$$

$$I = \frac{1}{3} * 0.186 * (3)^2 = 0.558slug/ft^2$$

$$L = 3ft \qquad m = \frac{6}{32.2} = 0.186$$

Steady-state Torque

$$Friction = \mu * N = 0.3 * 6 = 1.8lb$$

$$T_s = F * r = 1.8 * 2 = 3.6lb/in$$

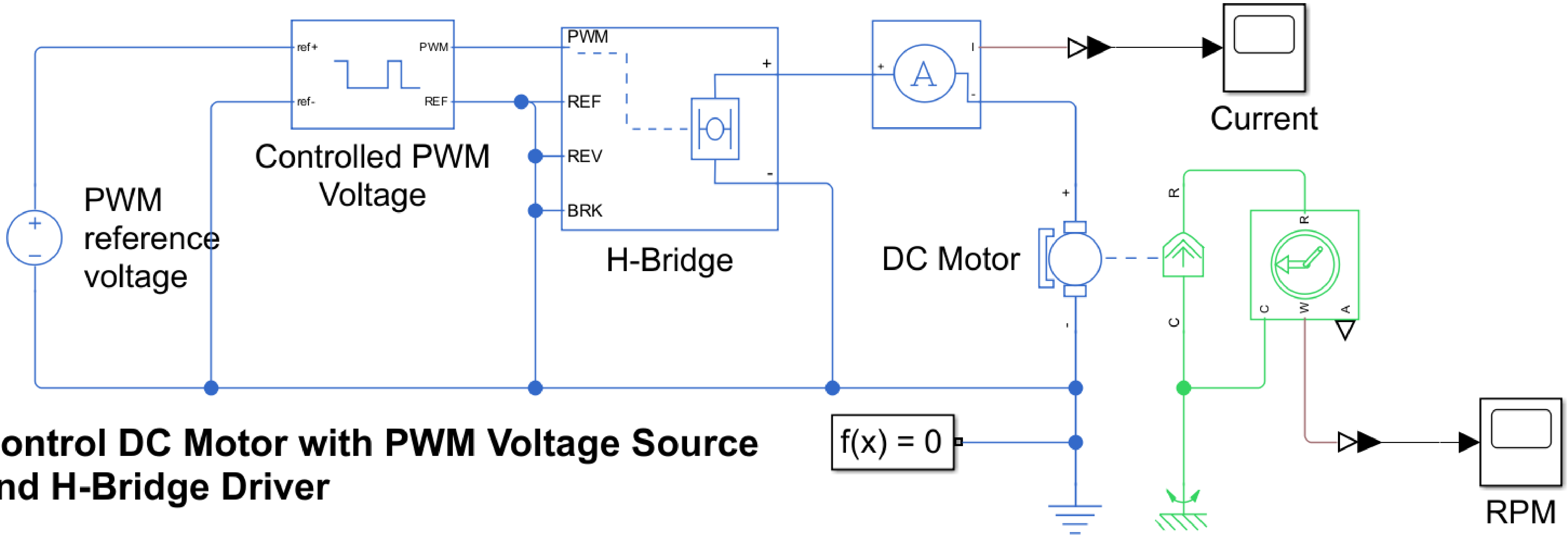
Torque-speed relationship

$$\frac{T_l load}{T_r rated} = 1 - \frac{N_l load}{N_n no - load}$$

$$\frac{T_l load}{8} = 1 - \frac{13}{25} = 0.48 \rightarrow T_l load = 0.48 * 8 = 3.84lb - in$$

Prepared by: Keldon Ngo
Presented by: Keldon Ngo

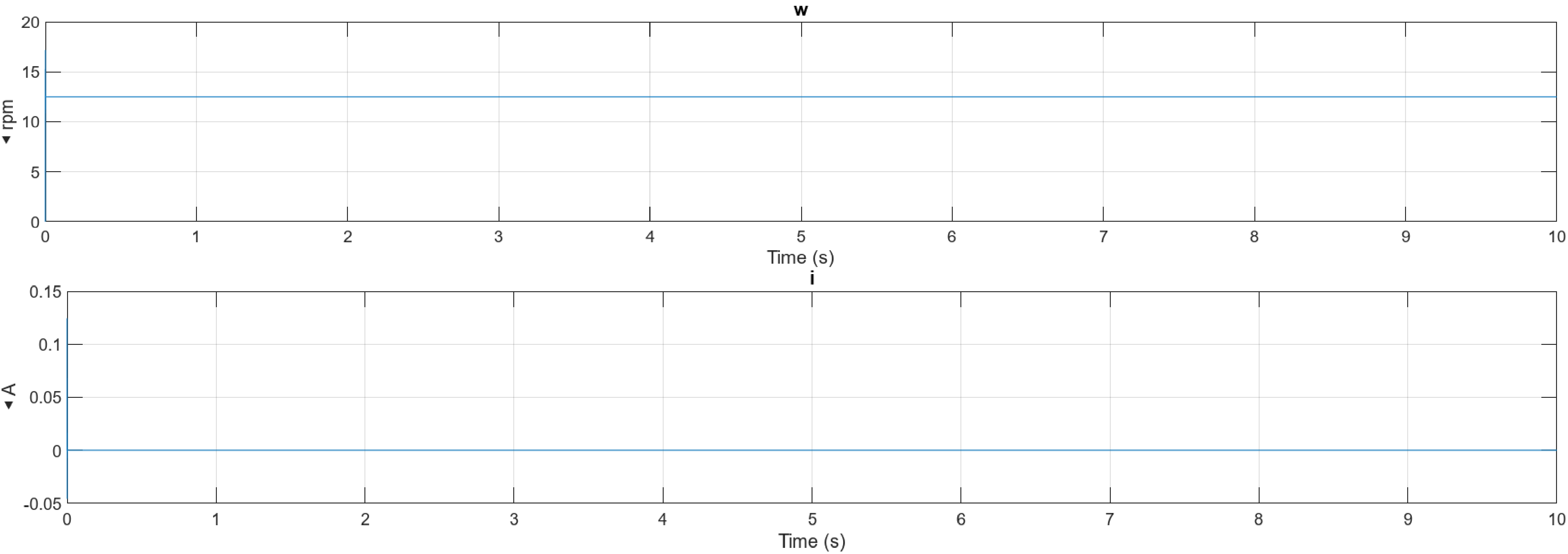
Engineering Analysis: DC Geared Motor



Control DC Motor with PWM Voltage Source and H-Bridge Driver

Prepared by: Keldon Ngo
Presented by: Keldon Ngo

Engineering Analysis: DC Geared Motor



Prepared by: Kenny Bui
Presented by: Kenny Bui

Engineering Analysis: Overall Circuit Load

Original Circuit Load Analysis

Load #	A*min	Ah
1	65.5	1.0916667
2	74.5	1.2416667
3	43.32	0.722
4	37.34	0.6223333
5	131	2.1833333
6	149	2.4833333
7	36.46	0.6076667
8	131	2.1833333
9	36.46	0.6076667
10	131	2.1833333
11	100.045	1.6674167
12	18.44	0.3073333
13-21	33.402	0.5567
22	48	0.8
23	150	2.5
Total		**19.757783**

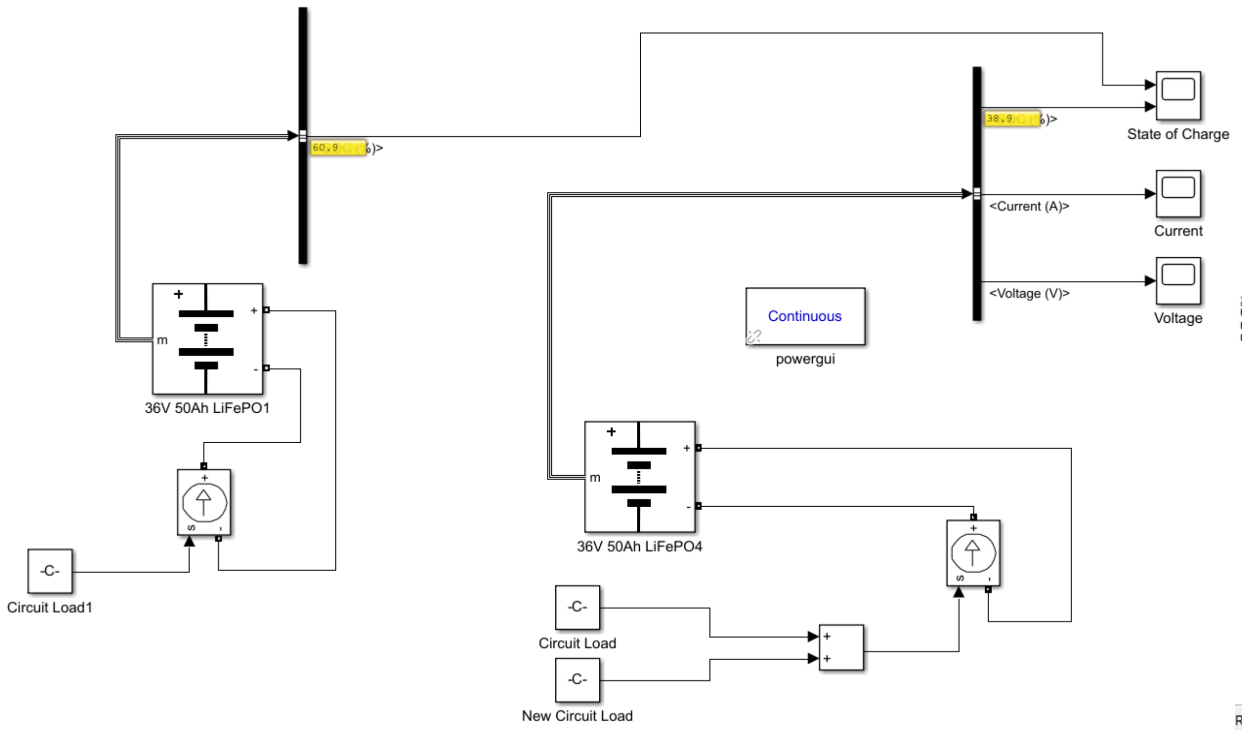
Load Calculation Conducted by Previous Group

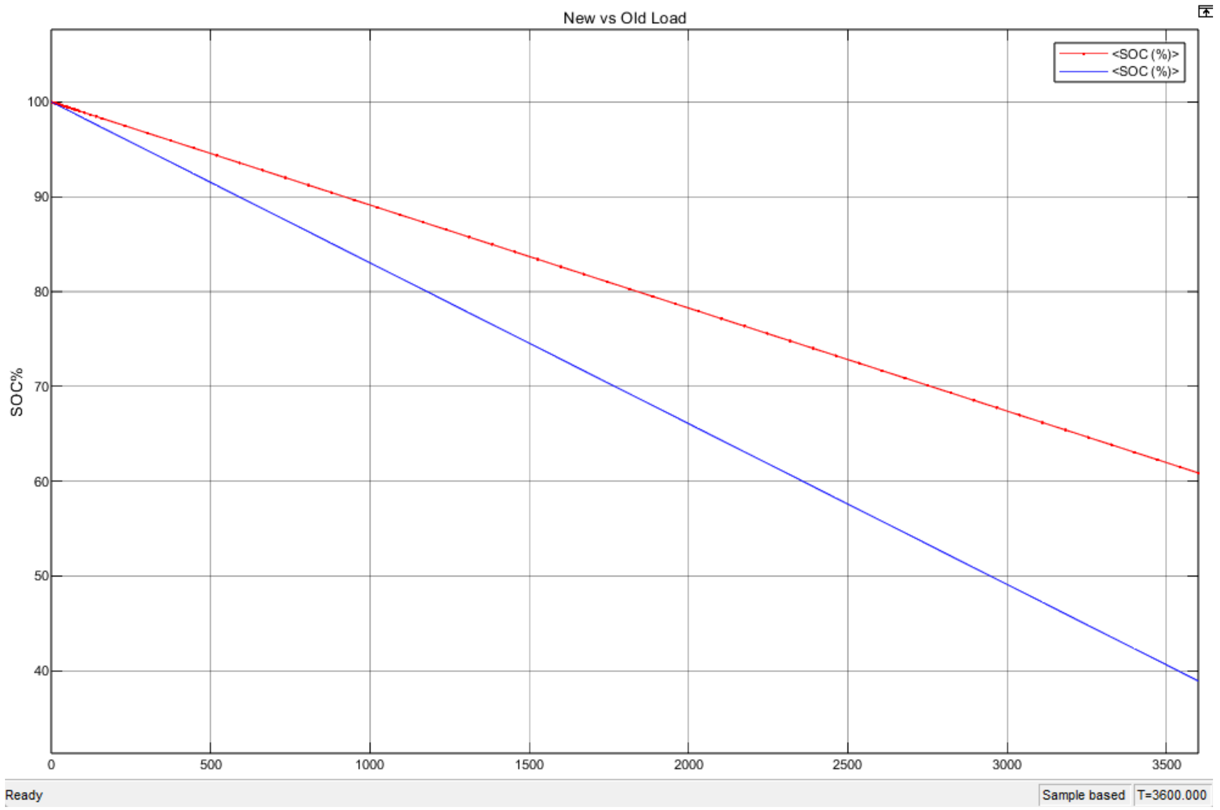
Additional Circuit Load Analysis

Load		A*min	Ah
Arduino Uno		0.0083	0.5
NEMA 23 Stepper Motors(Idle)		0.0555	3.33
NEMA 23 Stepper Motors(Running)		0.111	6.67
TB6600 Motor Drivers		0.0067	0.4
DC Brush Motor		0.002	0.125
L298N Motor Driver		0.0006	0.036
Charge Controller		0	0
MorningStar RD-1		0.0001	0.006
Buzzer		0.00025	0.015
			11.082

Prepared by: Kenny Bui
Presented by: Kenny Bui

Engineering Analysis: Overall Circuit Load





Red: Original Load (60.9%) Blue: New Load (38.9%)

Prepared by: Keldon Ngo
Presented by: Keldon Ngo

Component Selection: Solar Panel

REC Pure-RX Solar Panel

Power Output	450 W
Efficiency	21.6%
Power Density	20.1 W/m^2
Nominal Power Voltage	54.3 V
Nominal Power Current	8.29 A

REC
SOLAR'S MOST TRUSTED

Prepared by: Bowen Williamson
Presented by: Bowen Williamson

Component Selection: Solar Charge Controller

MorningStar TriStar-MPPT-30

Operating Voltage	9 V
Nominal Battery Voltage	12, 24, 36, or 48 V
Charge Rating	30 A
Max PV Power	1200 W
Max PV Voc	150 V

Prepared by: William Mancuso
Presented by: William Mancuso

Component Selection: Relay Driver

MorningStar RD-1

Operating Voltage	8-68V
Max Channel Current	750 mA
Voltage Accuracy	2% ± 50 mV
Transient Surge Protection	1500 W

Prepared by: Bowen Williamson
Presented by: Bowen Williamson

Component Selection: Pyranometer

Apogee Instruments SP-215-SS

Operating Voltage	5.5-24 V
Current Draw	300 uA
Output Voltage	0-5 V
Response Time	< 1 ms

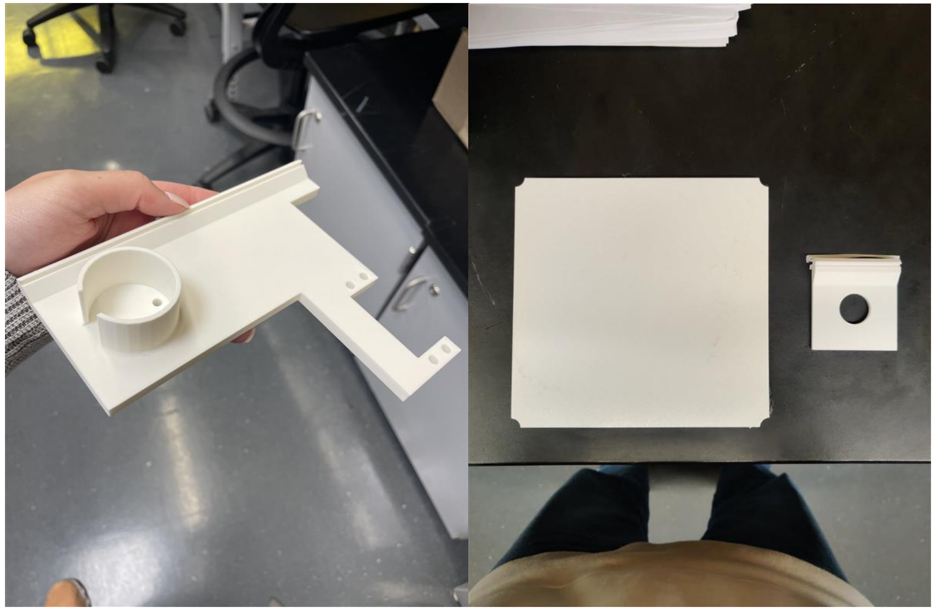
Prepared by: William Mancuso
Presented by: William Mancuso

Manufacturing Timeline

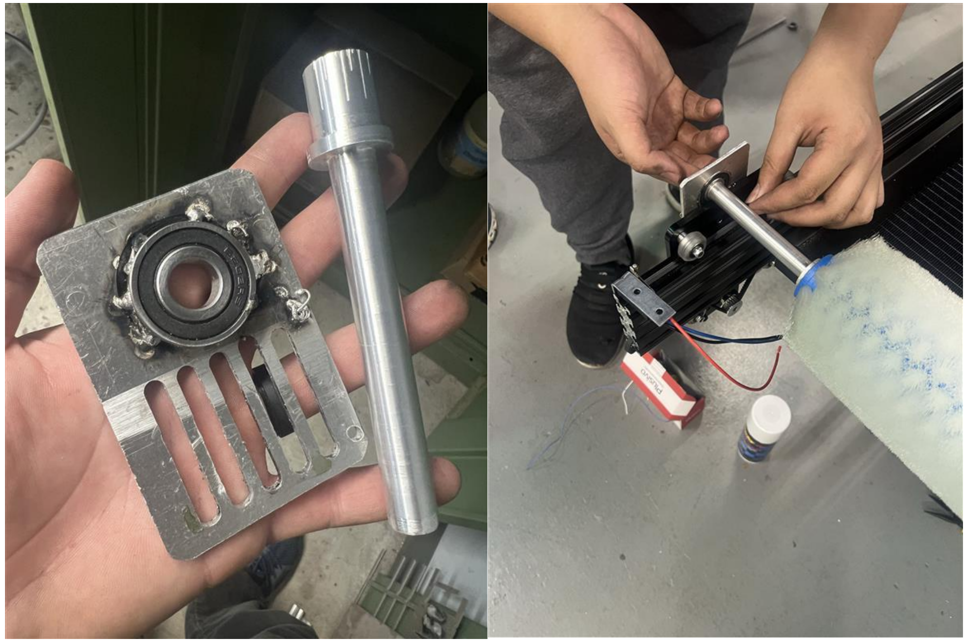
January	Manufacturing Plans and Order Parts
February	3D Prints (Parts and Stencils)
March	Frame Assembly and Machining Practice
April	Motor Housing and Connectors

Prepared by: William Mancuso
Presented by: William Mancuso

Manufacturing



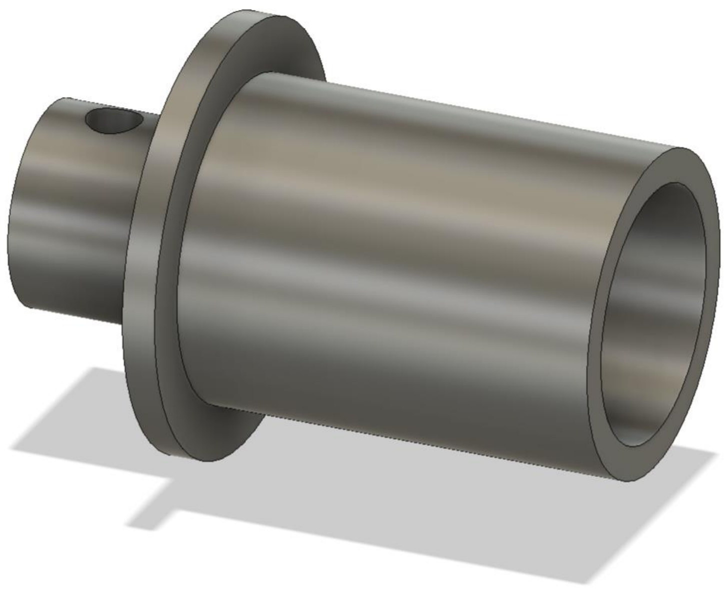
3D printed parts to hold components



Gantry Plates and Flange Couplings

Prepared by: William Mancuso
Presented by: William Mancuso

Manufacturing



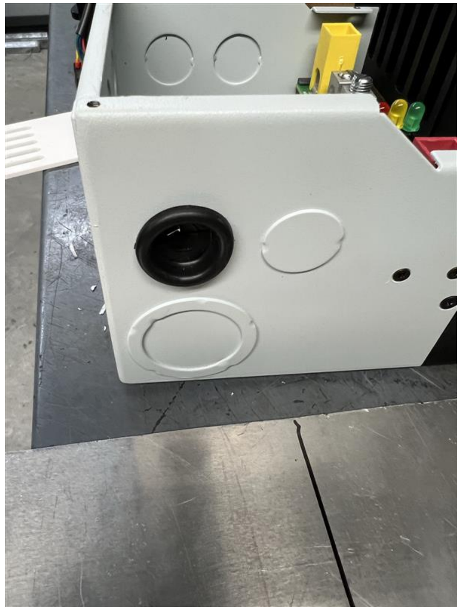
Flange coupling
3D model

Flange coupling
being turned on lathe

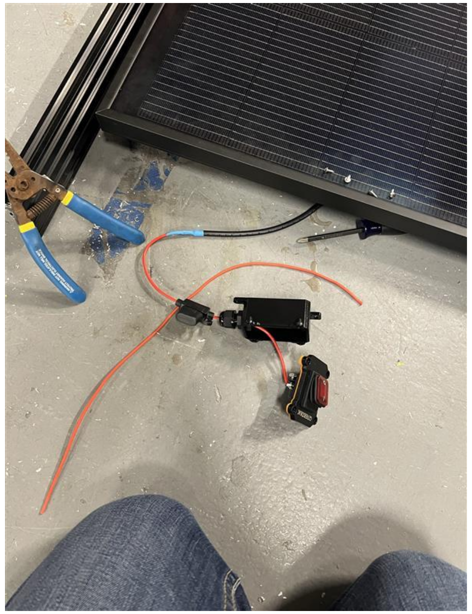
Welding of the t-track rails to the panel

Prepared by: William Mancuso
Presented by: William Mancuso

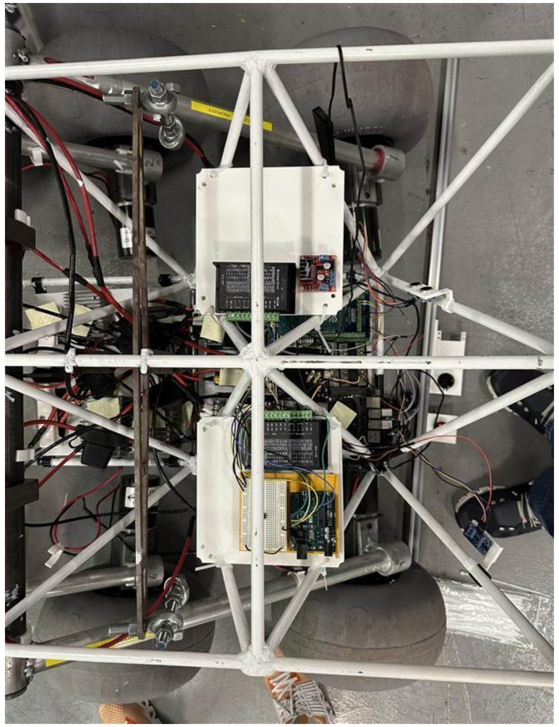
Manufacturing: Electrical

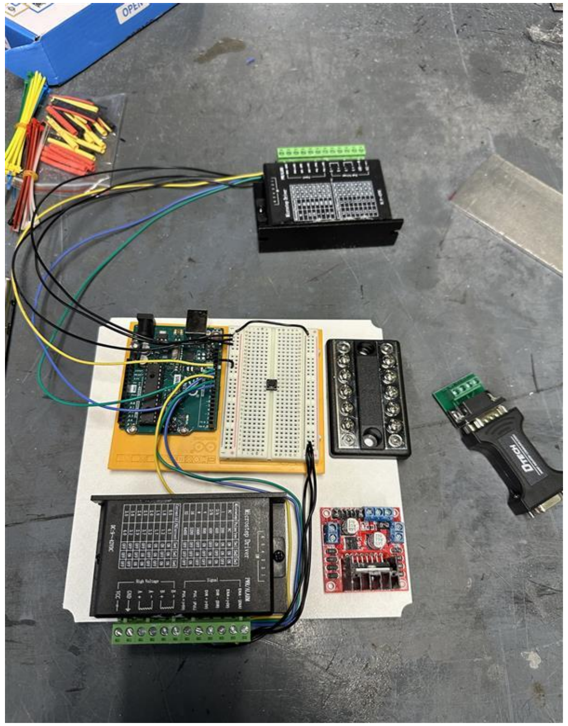


Wire grommets for charge controller



Removing the panel IP68 cable connector to add a switch





Electronics wiring and 3D printed plates for securing

Prepared by: William Mancuso
Presented by: William Mancuso

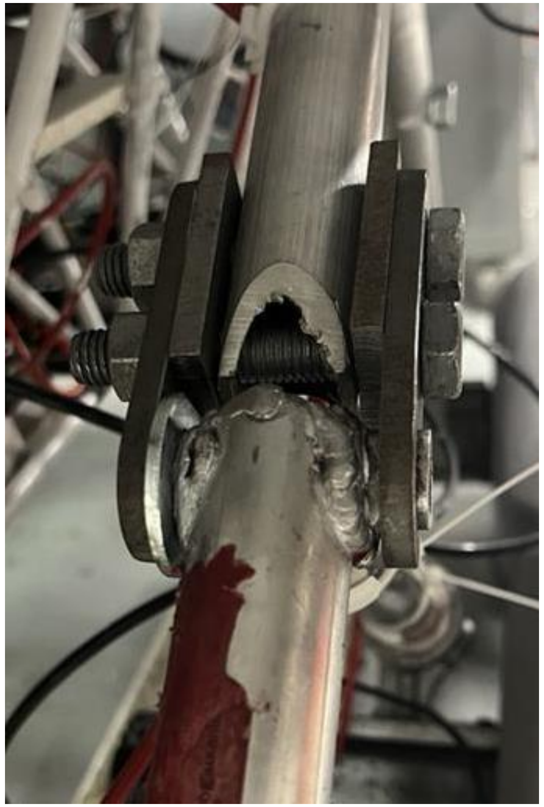
Manufacturing: Design Enhancements

Broken axle

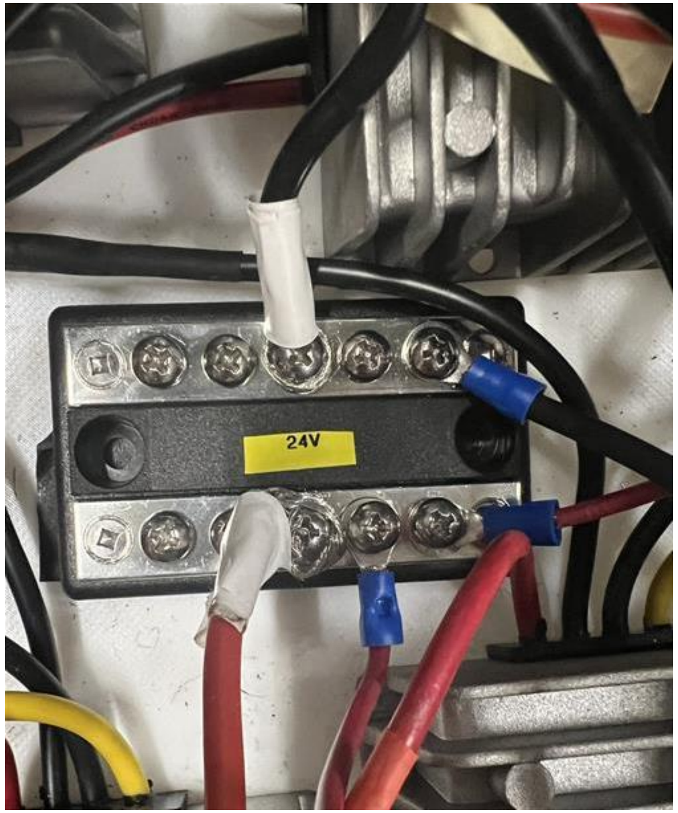
New and improved axle design

Prepared by: William Mancuso
Presented by: William Mancuso

Manufacturing: Design Enhancements



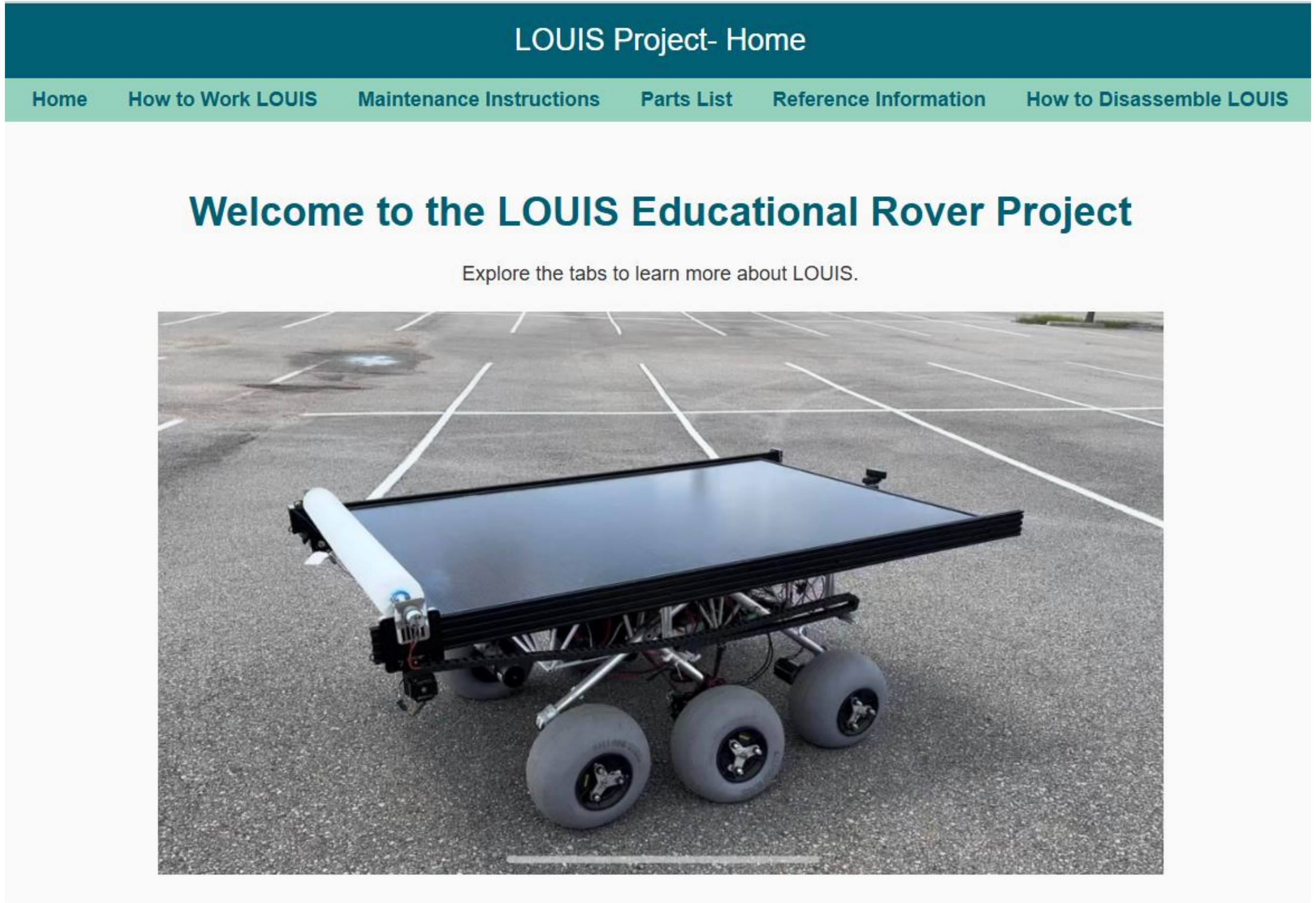
Washers on the rocker pivot to have a tighter fit



24V Bus bar and labels

Prepared by: Margaret Burkes
Presented by: Margaret Burkes

Manufacturing: Website



Prepared by: Keldon Ngo
Presented by: Keldon Ngo

Testing and Validation: Solar Panel

Sunny Condition Test:

Day:	Time:	Voltage:	Current:	Power:
2/25	11:33AM	50.1V	7.0 A	350.7 W
2/25	11:39AM	50.3V	7.1A	357.1 W
4/25	12:15PM	54.9V	8 A	439 W

Prepared by: Keldon Ngo
Presented by: Keldon Ngo

Testing and Validation: Solar Panel

Cloudy/Partly Cloudy Condition Test:

Day:	Time:	Voltage:	Current:	Power:
3/13	10:58AM	51.5V	8.3A	427.5 W
3/13	11:06AM	48.6V	7.8A	379.1 W
3/19	3:25PM	50.7V	8.3A	420.8 W
3/19	3:40PM	45.7V	7.3A	333.6 W

Prepared by: Keldon Ngo
Presented by: Keldon Ngo

Testing and Validation: Solar Panel

Indoor Condition Test:

Day:	Time:	Voltage:	Current:	Power:
3/18	11:18AM	47.6V	6.45A	307.0W
3/18 (wall facing)	11:30AM	21.9V	2.80A	61.3W
3/18	11:35AM	46.9V	6.23A	292.2W

Prepared by: Bowen Williamson
Presented by: Bowen Williamson

Testing and Validation: Charging and Battery Operation

	Battery % Stationary	Battery % Full-Speed	Battery Life Expectancy
No solar power		-21.34% per hr	4.69 hr
With solar power	+8% per hr	-6.95% per hr	14.39 hr

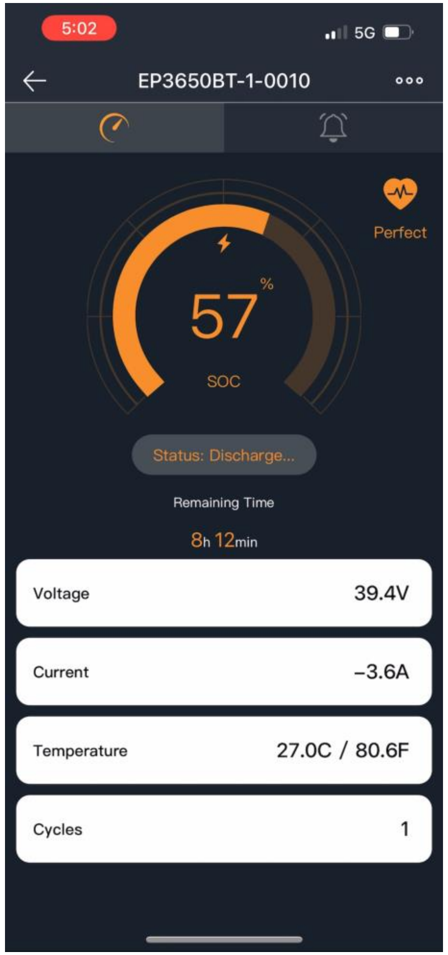
No Solar: $\frac{70\%}{3\text{hr } 17\text{min}} = \frac{70\%}{3.28\text{hr}} = -21.34\% \text{ per hour}$

Expected Life: $100\% \times \frac{1\text{hr}}{21.34\%} = 4.69 \text{ hours}$

With Solar: $\frac{57\%}{8\text{hr } 12\text{min}} = \frac{57\%}{8.2\text{hr}} = -6.95\% \text{ per hour}$

Expected Life: $100\% \times \frac{1\text{hr}}{6.95\%} = 14.39 \text{ hours}$

206.8% Increase

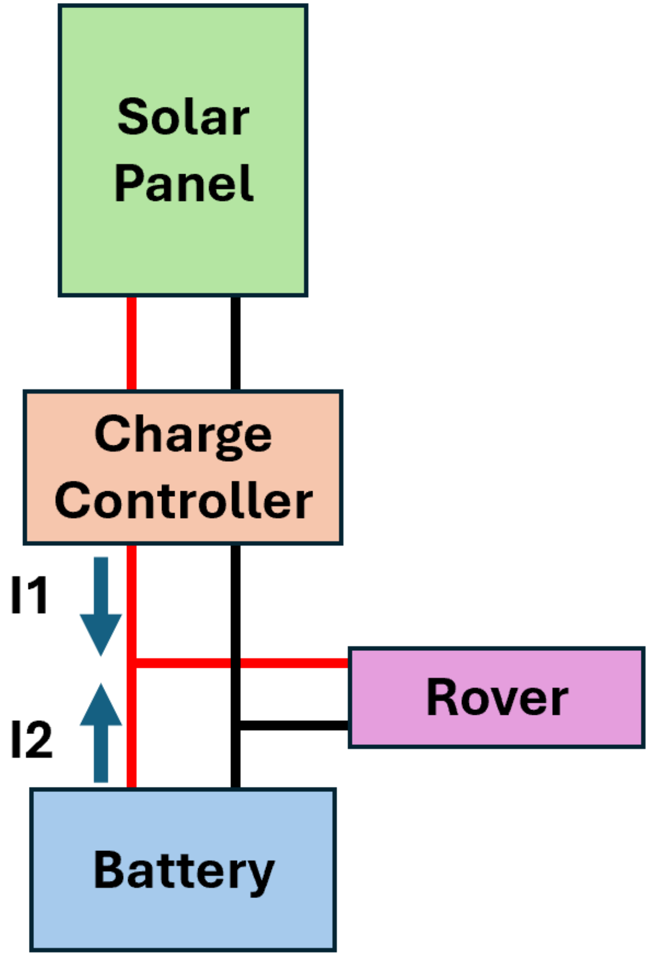


Prepared by: Bowen Williamson
Presented by: Bowen Williamson

Testing and Validation: Autonomous Power Flow

Full-Speed Current Measurements

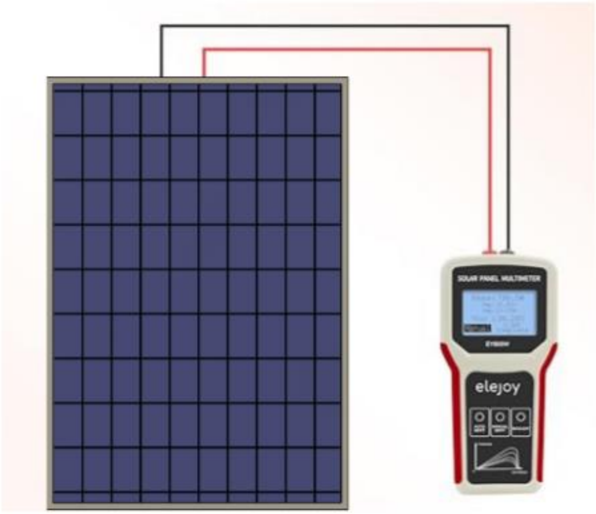
	No Solar	Cloudy	Full Sun
PV Current (I1)		2.5A	7.7A
Battery Current (I2)	11.0A	8.8A	3.4A



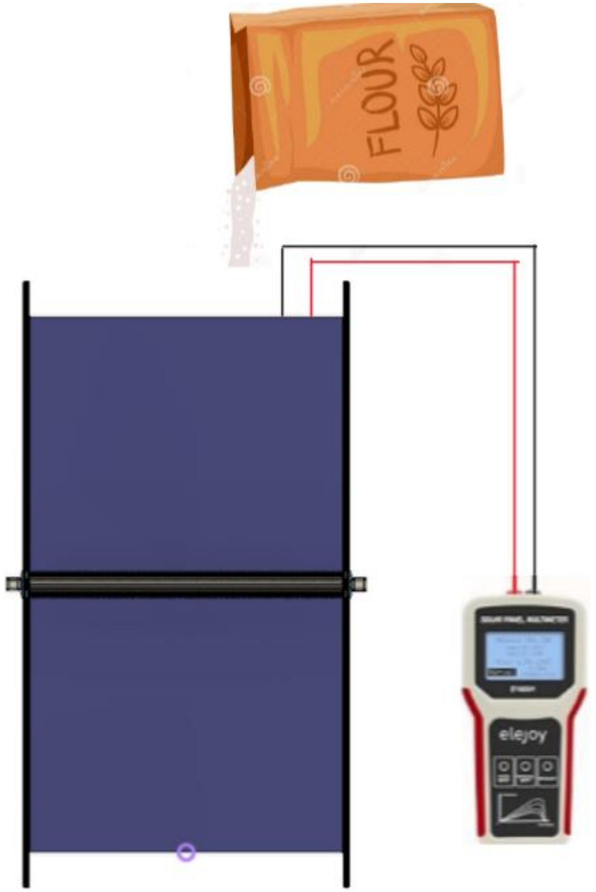
Prepared by: Margaret Burkes
Presented by: Margaret Burkes

Testing and Validation: Effectiveness of the Cleaning System

$$\eta_{cleaning} = \frac{P_{after} - P_{before}}{P_{before}} \times 100\%$$



Baseline Power:



Power After Cleaning:

Prepared by: Margaret Burkes
Presented by: Margaret Burkes

Testing and Validation: Effectiveness of the Cleaning System

	Current (A)	Power (W)
Before	8.00 A	450 W
During	2.25 A	125 W
After	6.5 A	350 W

180% improvement from dirty to clean

77.8% of Original Value

Prepared by: Kenny Bui
Presented by: Kenny Bui

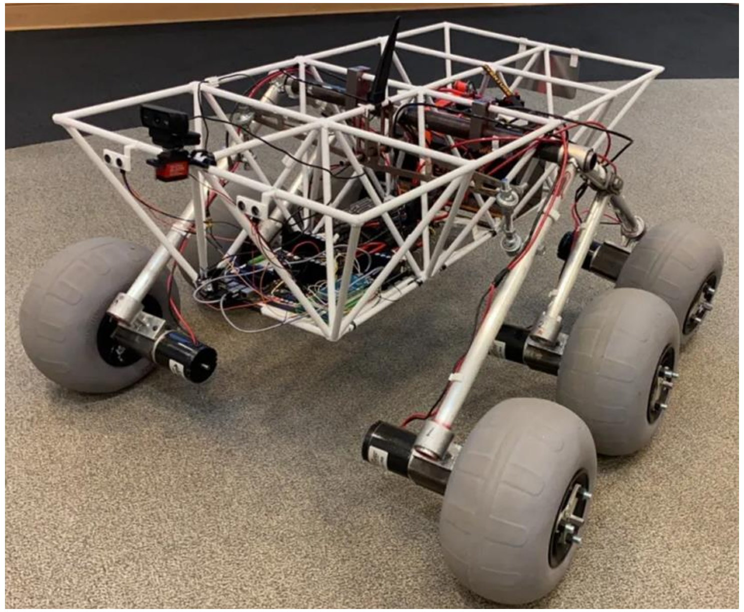
Testing and Validation: Activation of the Cleaning System

Prepared by: William Mancuso
Presented by: William Mancuso

Testing and Validation: Assembly Test

Assembly:
10 Minutes

Disassembly:
6.5 Minutes



Prepared by: William Mancuso
Presented by: William Mancuso

Testing and Validation: Weight, Dimensions, and Speed Test

	Original Results	**New Results**	**Difference**
Dimensions	1.25m x 1.25m x 0.625m	2.00m x 1.45m x 0.735m	+ 1.15 m^3
Weight	77.25 kg	113.4 kg	+36.2 kg
Speed	1.32 m/s	1.22 m/s	-0.1 m/s

Prepared by: Kenny Bui
Presented by: Kenny Bui

Safety Assessment

Concern	Safety Preventative
Electrical Shock Hazard	All electrical connections on our rover are properly grounded to prevent shock hazard.
Arc Flash Hazard	All electrical connections are covered and have appropriate safety measures in place to avoid short circuiting. This will mitigate arc flashes. Bus Bars have been labeled for ease of vision.
Overcurrent	All major electrical components have incorporated fused connections that are sized properly and are expected to pop in the case of overcurrent.
Emergency Stop	There is a cancel button designed in the brush cleaning system that is designed to cancel all operation of the cleaning system in case it is needed. Labeled for ease of vision.
Audible Warning Sound	There is also a buzzer designed to trigger before the cleaning system is activated to alert all nearby prior to the cleaning system activating.
Kill Switch	There are two switches attached to the battery that can be quickly disengaged to de-energize the entire rover in the case that it is needed

Prepared by: Bowen Williamson
Presented by: Bowen Williamson

Environmental Impact Assessment

- Decrease in power required from the grid
- Recyclable components
- Emission-free operation
- Testing with biodegradable materials (flour)
- Dry cleaning; no harmful chemicals

Prepared by: Bowen Williamson
Presented by: Bowen Williamson

Budget

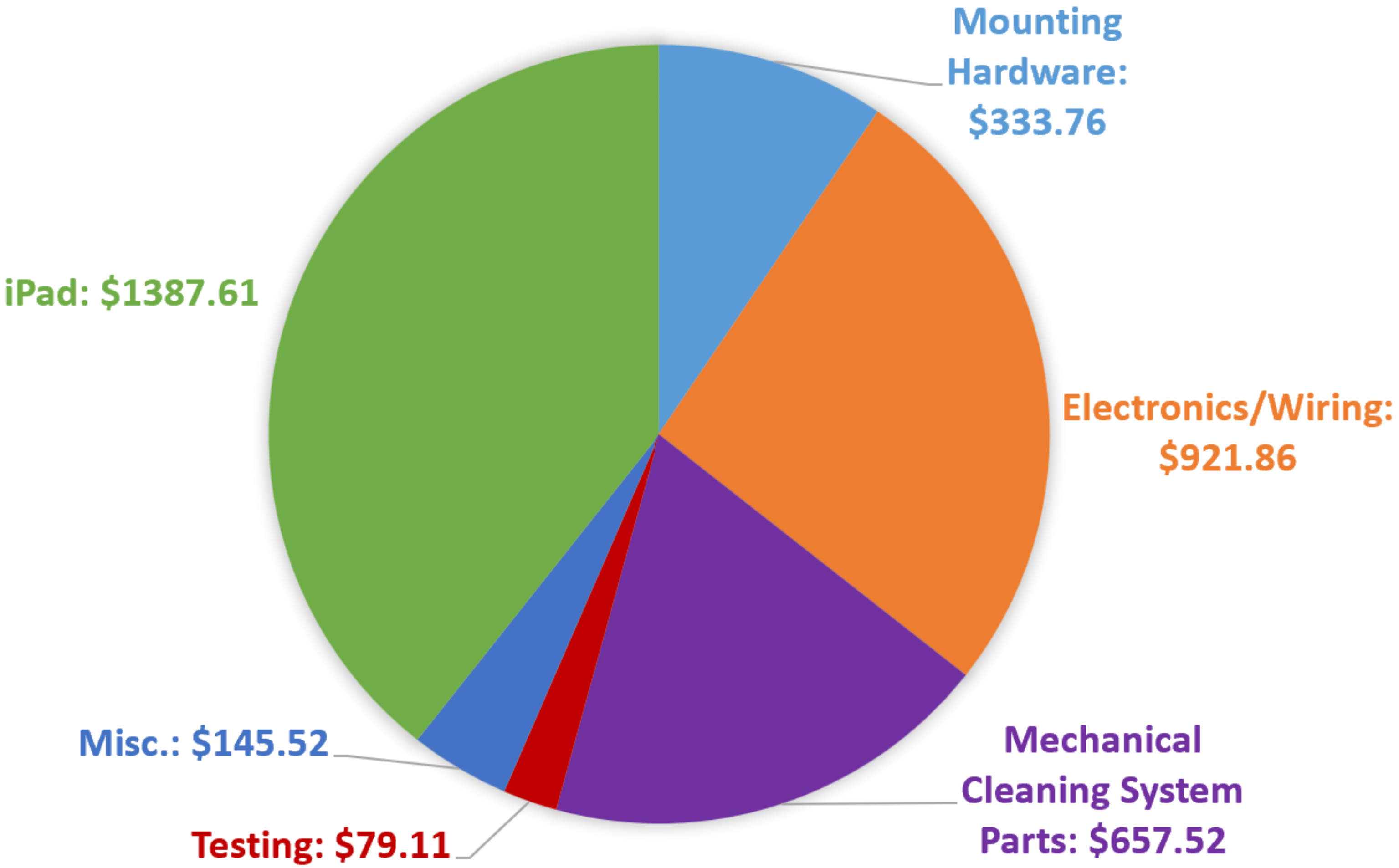
Allotted Budget: **$4000**

Total Cost: **$3525**

Generous Donations:

MorningStar
- TriStar Solar Charge Controller **($293)**

REC Solar
- 450W Pure-RX Solar Panel **($234)**



Prepared by: Kenny Bui
Presented by: Kenny Bui

Timeline:

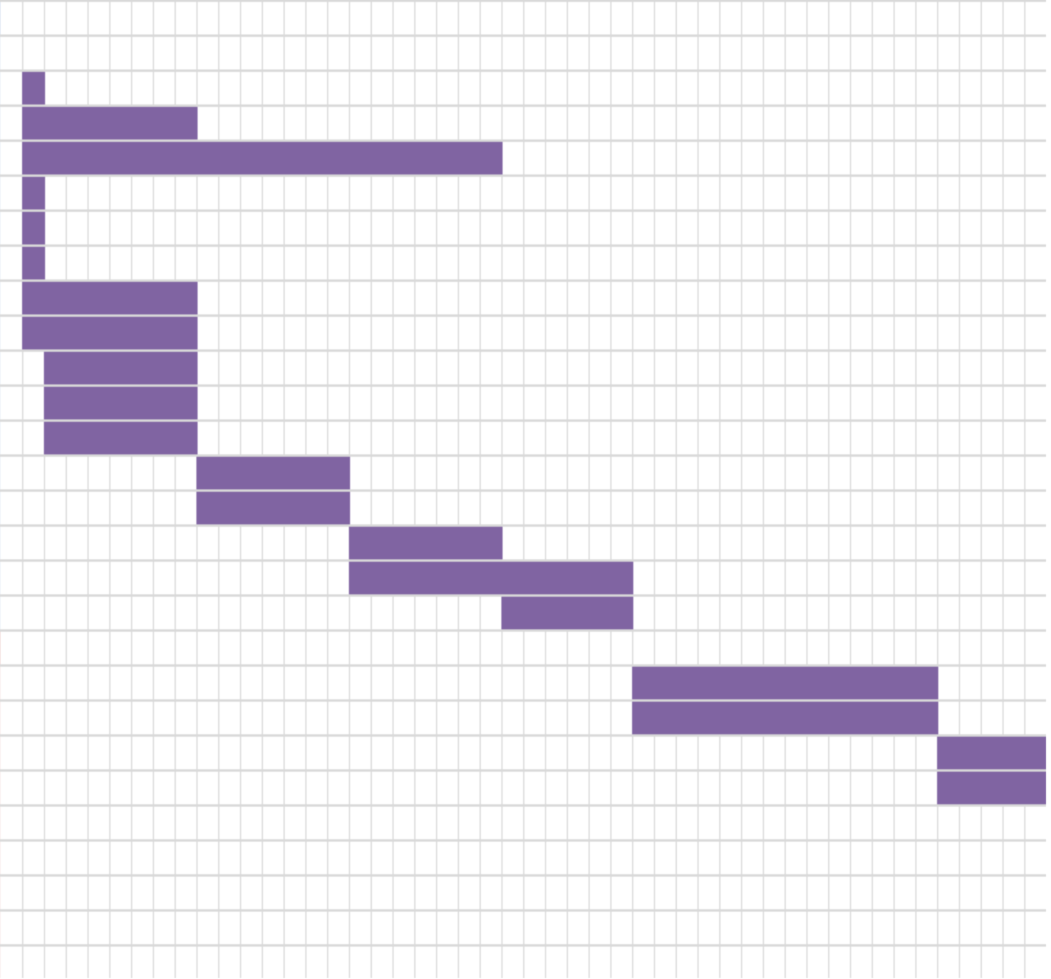
Project Preview			
Met Sponsor/Decide Faculty Advisor	100%	9/5/24	9/6/24
Project Interpretation and Team Roles	100%	9/7/24	9/10/24
Planning and Design			
Research Previous Rover Project	100%	9/11/24	9/12/24
Research Solar Panel Cleaning	100%	9/13/24	9/16/24
Research Methods to Autonomously Switch	100%	9/14/24	9/17/24
Research Solar Panels & Connections	100%	9/15/24	9/18/24
Create Concept Generations for Each Functio	100%	9/19/24	9/24/24
Decide/Select Ideal Concept Generations	100%	9/25/24	10/2/24
Create 3D Models in Fusion for Each Functio	100%	10/3/24	10/20/24

Analysis		
Simulations to See Power Capabilities of Solar Panel	10/21/24	10/28/24
Solar Panel with Rover System as Load Test	10/21/24	10/28/24
Run Analysis on Charge Controller for Power Switching	10/29/24	11/4/24
Analysis on DC Brush Motor for Cleaning Brush	10/29/24	11/4/24
MATLAB Stepper Motor Analysis	10/29/24	11/4/24
Construct Code and Analyze For Cleaning System Activatio	11/5/24	11/11/24
Experiment with Physical Model with Real Debris	11/5/24	11/11/24
Load Analysis for New Electrical Components	11/12/24	11/18/24
Evaluate Results and Adjust Concepts Accordingly	11/12/24	11/18/24
Verify Compatibility With Previous Rover System	11/19/24	11/25/24
Order Necessary Parts For Manufacturing	11/26/24	12/7/24

Prepared by: Kenny Bui
Presented by: Kenny Bui

Timeline Cont.:

Manufacturing		
William Will Fix Rover Breaker	1/21/25	1/27/25
Kenny Will Be Coding Arduino	1/21/25	2/3/25
Keldon Will Design Gantry Plate	1/21/25	2/10/25
Margaret Will Make/Configure Website	1/21/25	2/24/25
Margaret In Charge Of Stepper Driver Configuration	1/28/25	2/3/25
William Will Fix and Adjust Camera	1/28/25	2/3/25
Keldon In Charge Of Gear Driver Configuration	1/28/25	2/3/25
Kenny Will Construct Linear Actuator	1/28/25	2/10/25
Kenny Will Incorporate Limit Switches	1/28/25	2/10/25
Margaret Is In Charge Stepper Motor Wiring	2/4/25	2/10/25
Keldon Is In Charge Of Gear Motor Wiring	2/4/25	2/10/25
William and Bowen In Charge Of Pyranometer Platform	2/4/25	2/10/25
William and Bowen In Charge Of Program Charge Controller	2/11/25	2/17/25
William and Bowen In Charge Of Pyranometer Wiring	2/11/25	2/17/25
William and Bowen In Charge Of Connect Charge Controller	2/18/25	2/24/25
William Will Assemble Solar Panel and Frame	2/18/25	3/2/25
Bowen Will Assemble Flange and Brush System	2/25/25	3/2/25
Testing		
Keldon Will Solo Test Brush Cleaning System	3/3/25	3/16/25
Kenny Will Ensure Motors Actuate Properly	3/3/25	3/16/25
Keldon Will Solo Test Solar Panel Power Generation	3/17/25	3/30/25
Keldon Will Test Production In Different Scenarios	3/17/25	3/30/25
William and Bowen Solo Test Autonomous Power Switching	3/31/25	4/13/25
William and Bowen Test Power Switching with Different SOC	3/31/25	4/13/25
The Group Will Test Rover With Total Integration	4/14/25	4/20/25
The Group Will Make Adjustments as Needed	4/14/25	4/20/25
The Group Will Conduct a Series of Final Tests	4/21/25	4/27/25

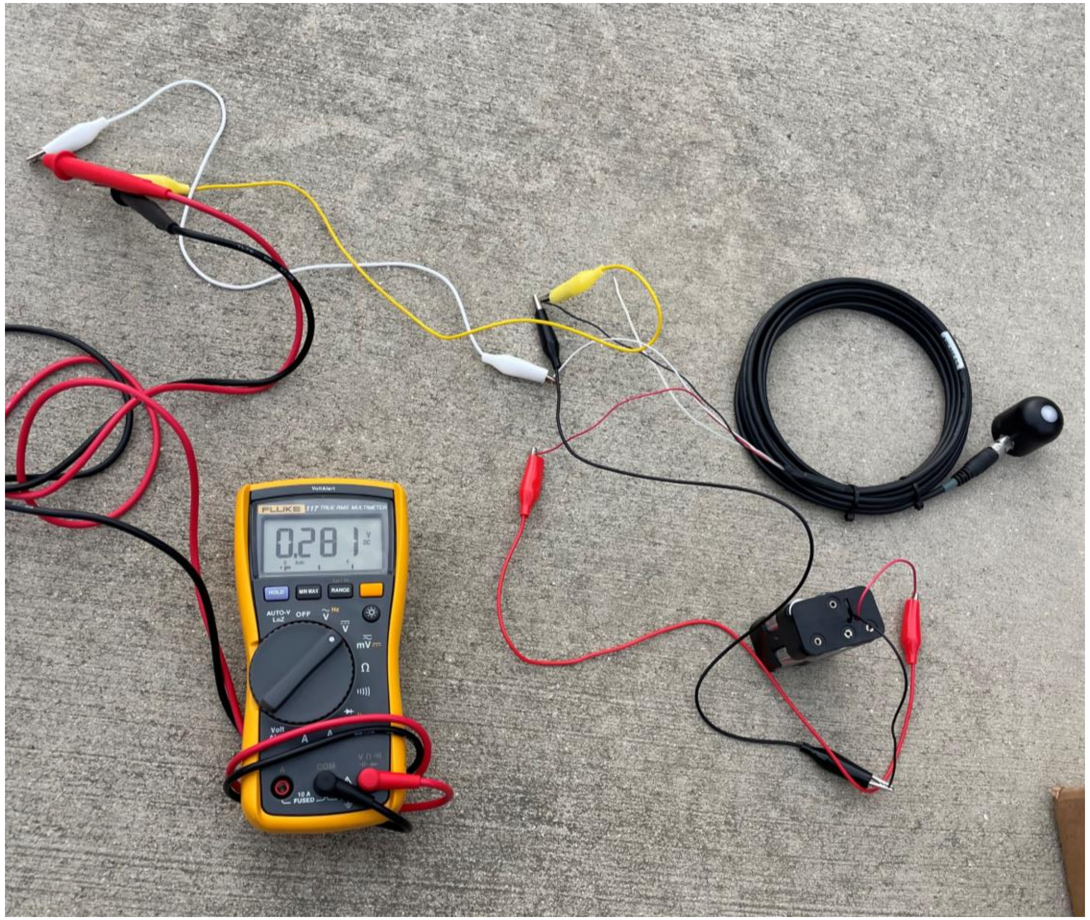


Appendix

Appendix

Prepared by: Margaret Burkes
Presented by: Margaret Burkes

Pyranometer Measuring Procedure



Prepared by: Margaret Burkes
Presented by: Margaret Burkes

Pyranometer Testing Results

3/13

Time:	Reading:	Weather:
10:56AM	1.60V	Partly Cloudy
10:57AM	1.717V	Partly Cloudy
10:59AM	1.714V	Partly Cloudy
11:00AM	1.770V	Partly Cloudy
11:01AM	1.777V	Partly Cloudy
11:02AM	1.270V	Partly Cloudy
11:03AM	0.498V	Shadow
11:05AM	0.742V	Cloudy
11:06AM	1.158V	Cloudy
11:08AM	1.941V	Partly Cloudy
11:09AM	1.950V	Sunny
11:11AM	2.010V	Sunny
11:11AM	0.685V	Shadow
11:34AM	0.003V	Indoor
11:36AM	0.004V	Indoor
11:36AM	0.000V	Shadow and Indoor

3/18

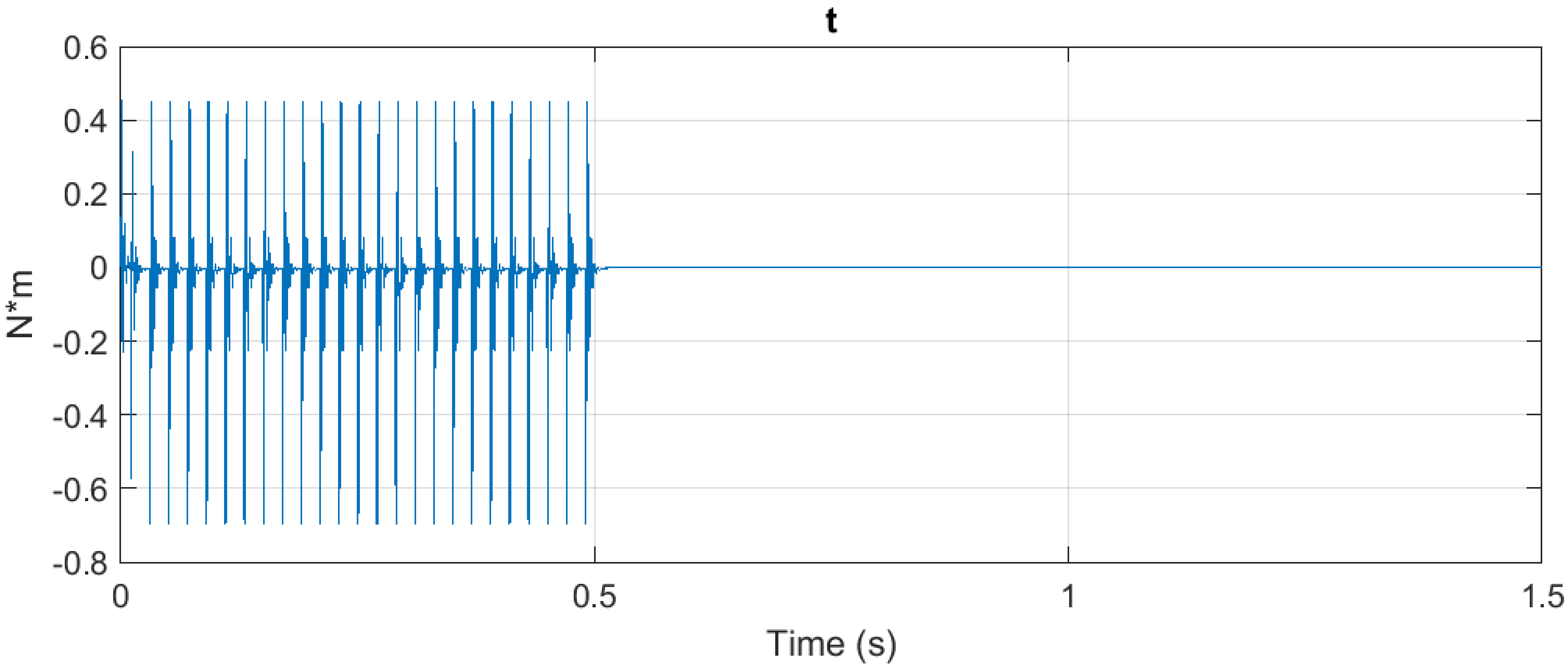
Time:	Reading:	Weather:
9:47AM	1.517V	Sunny
9:47AM	1.530V	Sunny
9:48AM	1.533V	Sunny
9:48AM	1.546V	Sunny
9:49AM	0.158V	Shadow
9:49AM	1.463V	Sunny
9:50AM	0.165V	Shadow
9:51AM	1.470V	Sunny
9:51AM	1.482V	Sunny
9:52AM	1.460V	Sunny
9:55AM	0.362V	Shadow
9:55AM	0.400V	Shadow
9:56AM	0.392V	Shadow
9:57AM	1.685V	Sunny
9:58AM	1.780V	Sunny
9:59AM	1.882V	Sunny

3/19

Time:	Reading:	Condition:
2:42PM	0.637V	Cloudy
2:42PM	0.627V	Cloudy
2:43PM	0.615V	Cloudy
2:43PM	0.592V	Cloudy
2:44PM	0.583V	Cloudy
5:49PM	0.328V	Partly Cloudy
5:50PM	0.299V	Partly Cloudy
5:50PM	0.242V	Partly Cloudy + Shadow
5:49PM	0.328V	Partly Cloudy
5:49PM	0.314V	Partly Cloudy
5:50PM	0.309V	Partly Cloudy
7:15PM	0.008V	Partly Cloudy
7:16PM	0.004V	Partly Cloudy and Shade

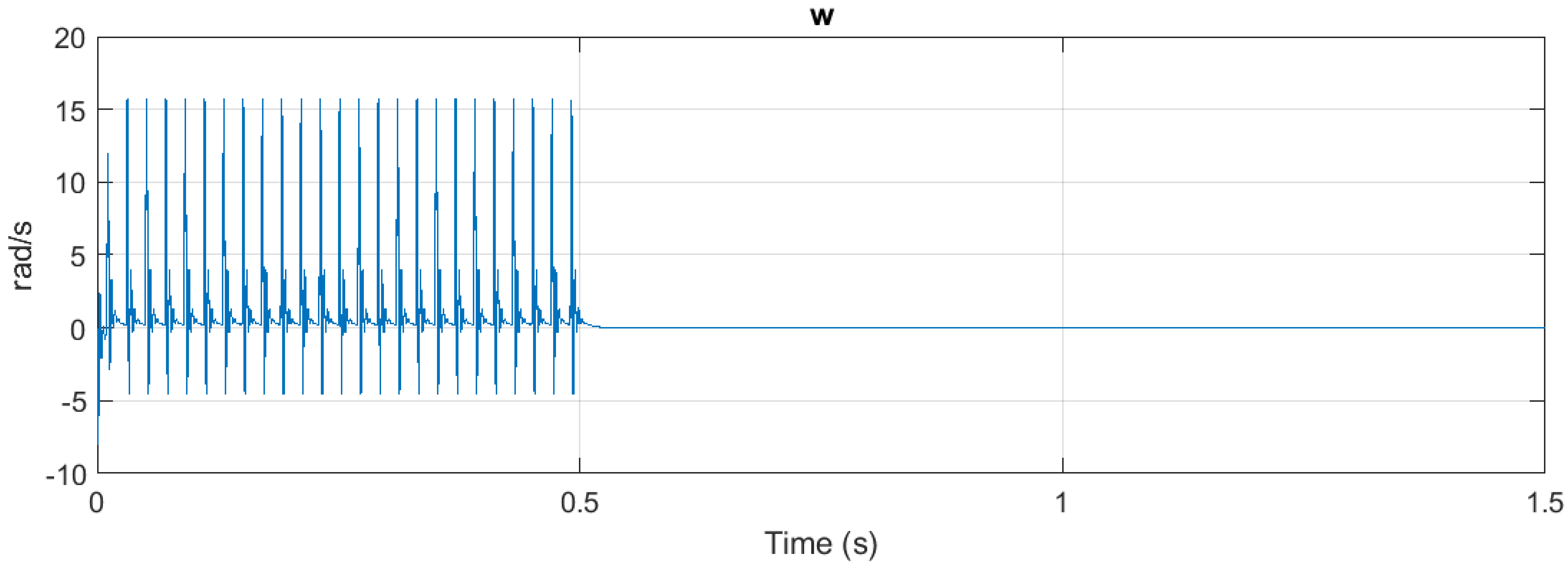
Prepared by: Margaret Burkes
Presented by: Margaret Burkes

Stepper Motor Torque



Prepared by: Margaret Burkes
Presented by: Margaret Burkes

Stepper Motor Speed



Prepared by: Margaret Burkes
Presented by: Margaret Burkes

Website-How to Work LOUIS

LOUIS Project - How to Work LOUIS

Home | How to Work LOUIS | Maintenance Instructions | Parts List | Reference Information | How to Disassemble LOUIS

How to Turn on LOUIS

The following guide will help you operate and understand LOUIS.

Step 1: Turn on Battery Power for the Charge Controller Switch

On the side of the battery, there is a switch with a battery icon. Turn this on to start.

Step 2: Turn on the Rocker Switch for the Solar Panel

On the side of the charge controller, this is a rocker switch. Turn this on.

Prepared by: Margaret Burkes
Presented by: Margaret Burkes

Website-Maintenance Instructions

LOUIS Project - Maintenance Instructions

Home | How to Work LOUIS | Maintenance Instructions | Parts List | Reference Information | How to Disassemble LOUIS

Maintenance Instructions

Brush

When to Replace:

- For reference, it was first used in February of 2025.
- As a general rule of thumb, the brush should be replaced every 6-12 months.
- **OR** if there are worn, damaged, or frayed bristles, consider replacing the brush to minimize damage to the solar panel.
- **OR** if the brush does not effectively remove debris, consider replacing the brush to maximize the solar panel's efficiency.

Prepared by: Margaret Burkes
Presented by: Margaret Burkes

Website-Parts List

Parts List

Search for parts...

Component	Manufacturer	Link	Extra Details
Tufted Brush	Industrial Brush Corp	None. You have to email the company for customized brushes.	.012 crimped natural 6-6 nylon 44" x 4" OD 18 row x 5/8" OC 44" OAL x 1-1/2" OD x 3/4" ID No end fittings
4 Gauge Wire	GearIT	View Product	25 feet each-black/red translucent
Waterproof 100Amp Circuit Breaker	RED WOLF	View Product	
TB6600 Stepper Motor Driver	OUYZGIA	View Product	For NEMA 23 stepper motor
Arduino Uno Rev3	Arduino	View Product	
L298N Motor Driver	Generic	View Product	

Prepared by: Margaret Burkes
Presented by: Margaret Burkes

Website-Reference Information

Reference Info

Explore the PowerPoints and other documents shown below for more background information.

Project #57

CPFR:

Download PDF

FEMD:

Download PDF

CPFPS:

Download PDF

CPP:

Download PDF

Project #66

EATB:

Download PDF

FDPR:

Download PDF

ESIR:

Download PDF

PiTA:

Download PDF

Battery Monitoring

Monitor the Enduro 36V battery life with the Enduro Power App.

Open Enduro Power App

Prepared by: Keldon Ngo
Presented by: Keldon Ngo

Website- How to Disassemble LOUIS

LOUIS Project - How to Disassemble LOUIS

Home | How to Work LOUIS | Maintenance Instructions | Parts List | Reference Information | How to Disassemble LOUIS

How to Disassemble LOUIS

The following guide will help you disassemble LOUIS for transportation purposes. The disassembly takes around 6.5 minutes, while the assembly time takes around 10 minutes.

The rover's fully assembled state has the solar panel charge controller system fully connected and the cleaning system fully connected to the arduino. A detailed wiring diagram will also be included to make sure all the wires are correctly reassembled. Before doing any work make sure all power is off.

Step 1: Unscrew Electrical Conduit Straps

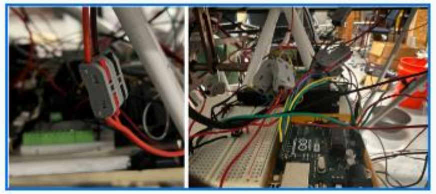
To begin disassembly, first start by unscrewing all of the electrical conduit straps from underneath the panel and frame. There are 4 straps with 2 screws each.

Note: Having partners for the next portion of the disassembly can make the job much easier by having them lift the panel slightly to get to any hard to reach screws.

Prepared by: Keldon Ngo
Presented by: Keldon Ngo

Website- How to Disassemble LOUIS Pt 2.

Step 2: Disconnect Quick Connections



The next step, move to the front of the rover, and begin disconnecting all wiring to the stepper motors and rotational motors, using their quick connections to their motor driver. These clips make it easier to unwire and rewire the small connections. This also goes for the connections to the limit switches in the arduino.

Step 3: Unscrew Charge Controller Face and Panel Wires

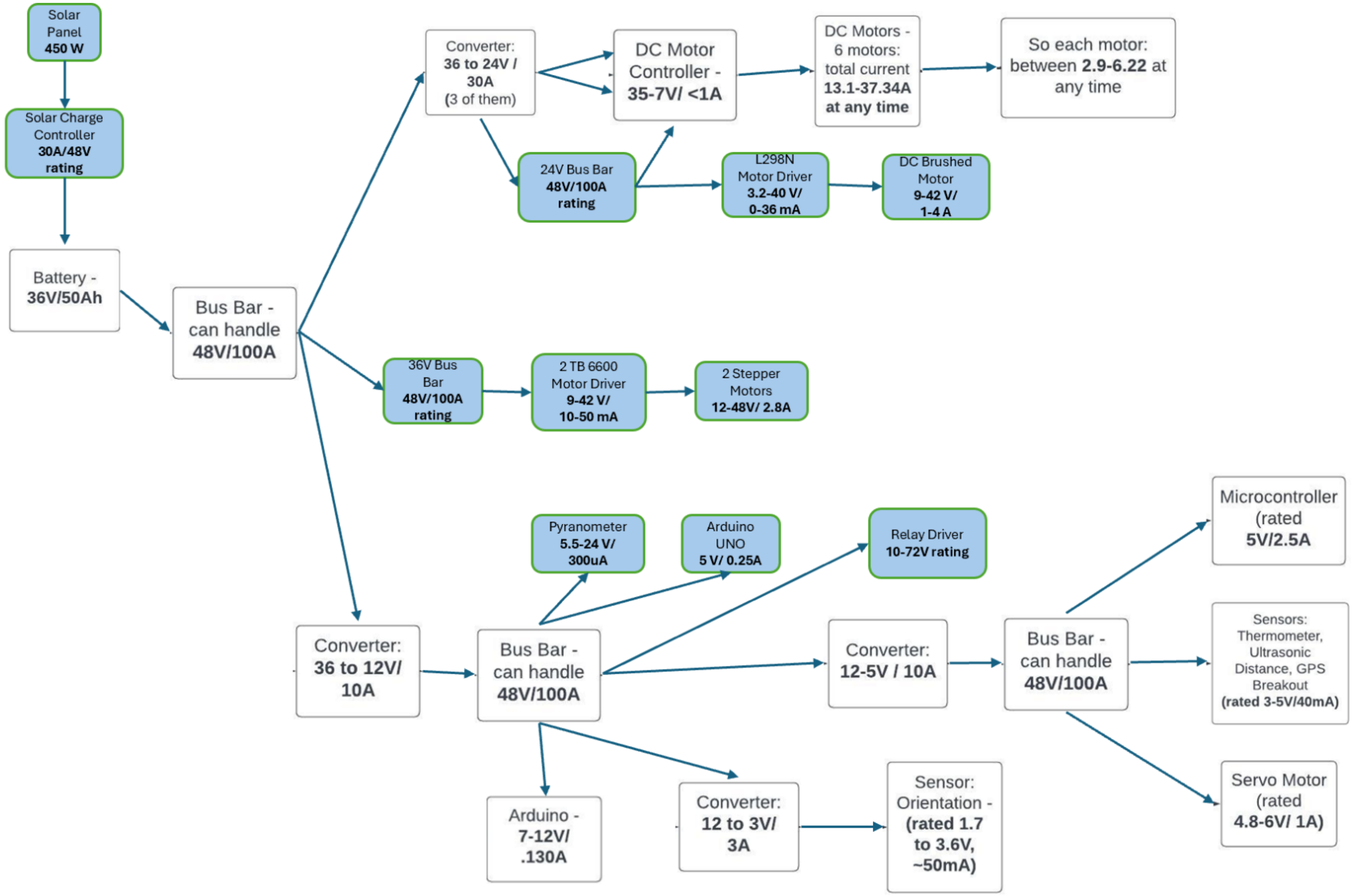
Finally move to the back of the rover to find the solar charge controller and begin removing the 4 screws from the face of the controller. Once that is removed, unscrew the panel wires connected to the points labeled array, and the ground. (The PV wires are the thin ones, the thick battery wires can be left alone.)

Step 4: Remove the Panel

After all this is complete, the panel can be lifted and removed by two people. For the safety of the equipment, it is recommended that the panel be lifted by the spots where the underneath t-track rails and the linear actuator rails meet. It should also be set down on the cable carrier L-Bars to protect the panel.

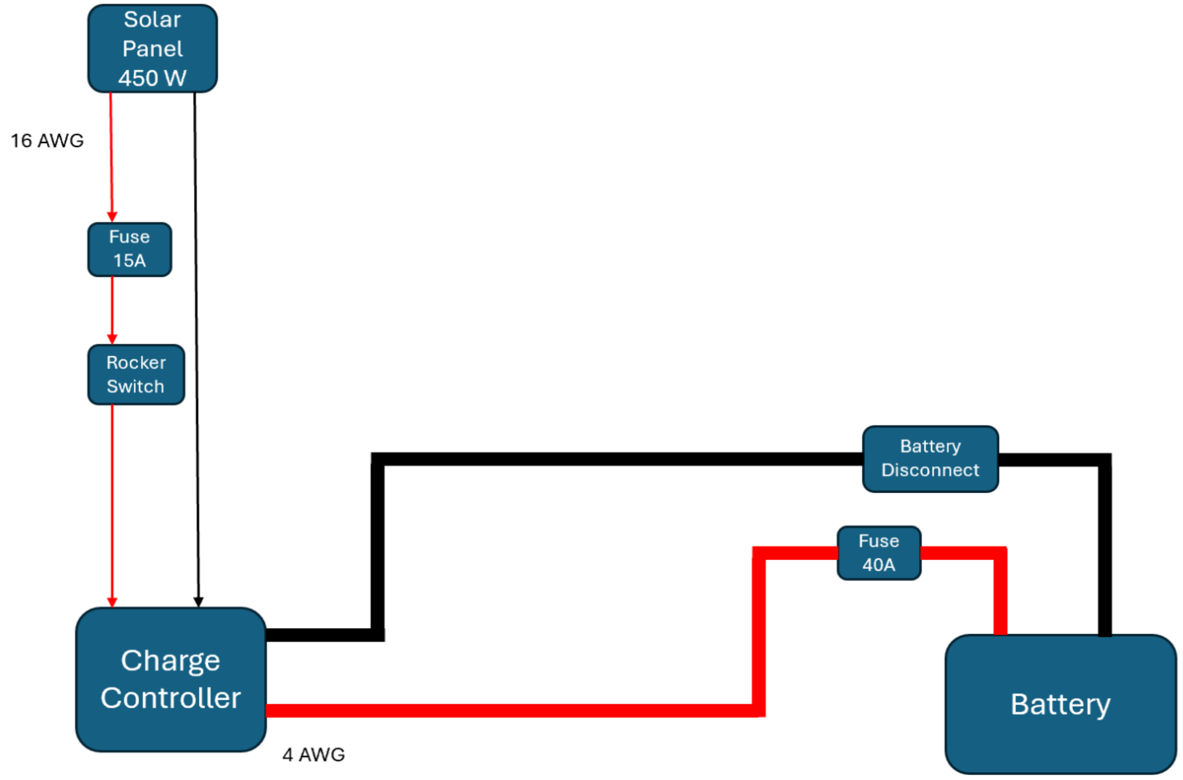
Prepared by: Kenny Bui
Presented by: Kenny Bui

Wiring Diagram – Overall Circuit



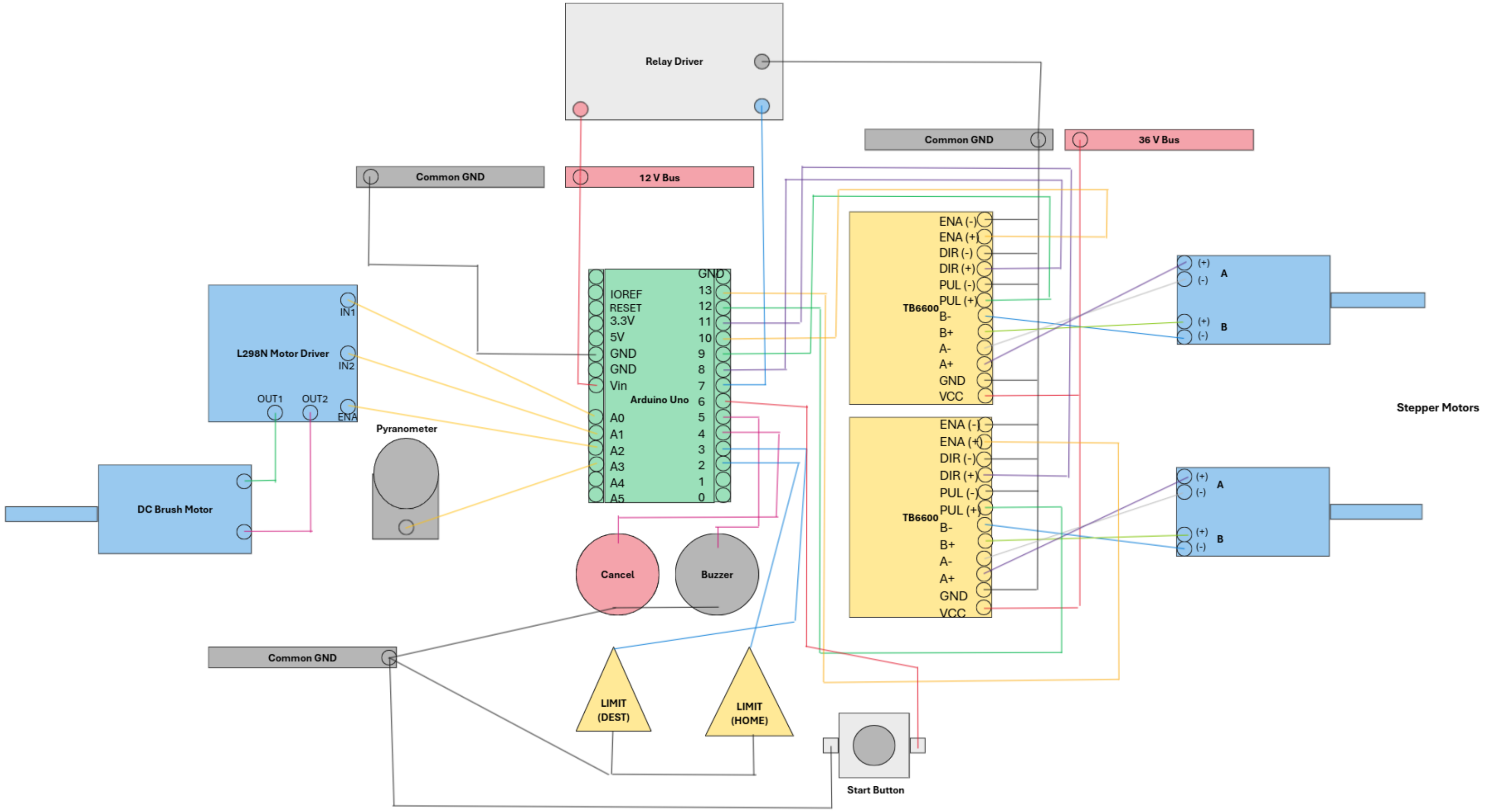
Prepared by: Kenny Bui
Presented by: Kenny Bui

Wiring Diagram – Solar Power System



Prepared by: Kenny Bui
Presented by: Kenny Bui

Wiring Diagram – Electronics



Prepared by: Kenny Bui
Presented by: Kenny Bui

Engineering Analysis: Cleaning Activation

```
// Stepper 1
#define DIR_PIN 8
#define STEP_PIN 9
#define ENABLE_PIN 10

// Stepper 2
#define DIR2_PIN 11
#define STEP2_PIN 12
#define ENABLE2_PIN 13

// Control buttons and sensors
#define LIMIT_HOME 2
#define LIMIT_DEST 3
#define CANCEL_BUTTON 4
#define BUZZER 5
#define START_BUTTON 6
#define RELAY_PIN 7

// Brush Motor (L298N)
#define BRUSH_IN1 A0
#define BRUSH_IN2 A1
#define BRUSH_EN A2

// Pyranometer (SP-215-SS)
#define PYRA_PIN A3
#define PYRA_THRESHOLD 300
#define ADC_TO_VOLTAGE 0.0048876
#define VOLTAGE_TO_Wm2 250

// Ramping config
#define STEP_DELAY_START 1000
#define STEP_DELAY_MIN 200
#define STEP_RAMP_STEPS 300

// Auto trigger hold timer
unsigned long autoTriggerStartTime = 0;
const unsigned long AUTO_TRIGGER_DELAY = 60000;  /

bool cancelTriggered = false;


void moveToLimit(int limitSwitch, bool direction);
void returnHome();
void stepMotors(int delayMicros);
void enableMotors();
void stopMotors();
void startBrushMotor(bool forward = true);
void stopBrushMotor();
void buzzerAlert();
void emergencyStopAndWait();
void runCleaningCycle();

void setup() {
  Serial.begin(9600);

  // Stepper pins
  pinMode(DIR_PIN, OUTPUT); pinMode(STEP_PIN, OUTPUT); pinMode(ENABLE_PIN, OUTPUT);
  pinMode(DIR2_PIN, OUTPUT); pinMode(STEP2_PIN, OUTPUT); pinMode(ENABLE2_PIN, OUTPUT);
  digitalWrite(ENABLE_PIN, HIGH); digitalWrite(ENABLE2_PIN, HIGH);

  // Brush motor
  pinMode(BRUSH_IN1, OUTPUT); pinMode(BRUSH_IN2, OUTPUT); pinMode(BRUSH_EN, OUTPUT);
  stopBrushMotor();

  // Inputs
  pinMode(LIMIT_HOME, INPUT_PULLUP);
  pinMode(LIMIT_DEST, INPUT_PULLUP);
  pinMode(CANCEL_BUTTON, INPUT_PULLUP);
  pinMode(START_BUTTON, INPUT_PULLUP);
  pinMode(RELAY_PIN, INPUT_PULLUP);
  pinMode(PYRA_PIN, INPUT);

  pinMode(BUZZER, OUTPUT); digitalWrite(BUZZER, LOW);

  Serial.println("System ready. Monitoring for manual or automatic cleaning...");
}
```

Prepared by: Kenny Bui
Presented by: Kenny Bui

Engineering Analysis: Cleaning Activation

```
void loop() {
  int relayState = digitalRead(RELAY_PIN);
  int pyranometerRaw = analogRead(PYRA_PIN);
  float pyranometerVoltage = pyranometerRaw * ADC_TO_VOLTAGE;
  float irradianceWm2 = pyranometerVoltage * VOLTAGE_TO_Wm2;
  bool manualStartTriggered = digitalRead(START_BUTTON) == LOW;

  Serial.print("Pyranometer: ");
  Serial.print(pyranometerRaw);
  Serial.print(" (");
  Serial.print(irradianceWm2);
  Serial.println(" W/m²)");

  //  Manual Start (instant)
  if (manualStartTriggered) {
    Serial.println("Manual START button pressed. Beginning cleaning...");
    while (digitalRead(START_BUTTON) == LOW);  // Debounce
    runCleaningCycle();
    return;
```

```
  // Auto Start (trigger when relay is LOW)
  if (relayState == LOW && pyranometerRaw >= PYRA_THRESHOLD) {
    if (autoTriggerStartTime == 0) {
      autoTriggerStartTime = millis();
      Serial.println("Auto trigger detected. Holding for 30 seconds...");
    } else if (millis() - autoTriggerStartTime >= AUTO_TRIGGER_DELAY) {
      Serial.println("Auto trigger held 30 seconds. Starting cleaning...");
      runCleaningCycle();
      autoTriggerStartTime = 0;
      return;
    } else {
      Serial.print("Auto trigger duration: ");
      Serial.print((millis() - autoTriggerStartTime) / 1000);
      Serial.println(" sec");
    }
  } else {
    if (autoTriggerStartTime > 0) {
      Serial.println("Auto trigger reset (conditions dropped).");
      autoTriggerStartTime = 0;
    }
  }

  delay(1000);
```

```
void runCleaningCycle() {
  cancelTriggered = false;
  enableMotors();
  buzzerAlert();

  moveToLimit(LIMIT_DEST, HIGH);
  if (cancelTriggered) return;

  delay(1000);
  returnHome();
  if (cancelTriggered) return;

  stopMotors();
  delay(5000);
  enableMotors();
}
```

Prepared by: Kenny Bui
Presented by: Kenny Bui

Engineering Analysis: Cleaning Activation

```
void moveToLimit(int limitSwitch, bool direction) {
  digitalWrite(DIR_PIN, direction);
  digitalWrite(DIR2_PIN, !direction);
  startBrushMotor(true);

  int stepCount = 0;
  while (digitalRead(limitSwitch) == HIGH) {
    if (digitalRead(CANCEL_BUTTON) == LOW) {
      Serial.println("Cancel pressed: stopping...");
      stopBrushMotor();
      stopMotors();
      cancelTriggered = true;

      Serial.println("Press START to return home...");
      while (digitalRead(START_BUTTON) == HIGH);
      while (digitalRead(START_BUTTON) == LOW);

      returnHome();
      return;
    }

    int delayMicros = STEP_DELAY_START;
    if (stepCount < STEP_RAMP_STEPS) {
      delayMicros = STEP_DELAY_START - ((STEP_DELAY_START - STEP_DELAY_MIN) * stepCount / STEP_RAMP_STEPS);
    } else {
      delayMicros = STEP_DELAY_MIN;
    }

    stepMotors(delayMicros);
    stepCount++;
  }

  stopBrushMotor();
```

```
void returnHome() {
  digitalWrite(DIR_PIN, LOW);
  digitalWrite(DIR2_PIN, HIGH);
  enableMotors();
  startBrushMotor(false);

  int stepCount = 0;
  while (digitalRead(LIMIT_HOME) == HIGH) {
    int delayMicros = STEP_DELAY_START;
    if (stepCount < STEP_RAMP_STEPS) {
      delayMicros = STEP_DELAY_START - ((STEP_DELAY_START - STEP_DELAY_MIN) * stepCount / STEP_RAMP_STEPS);
    } else {
      delayMicros = STEP_DELAY_MIN;
    }

    stepMotors(delayMicros);
    stepCount++;
  }

  stopBrushMotor();
  stopMotors();
}

void stepMotors(int delayMicros) {
  digitalWrite(STEP_PIN, HIGH); digitalWrite(STEP2_PIN, HIGH);
  delayMicroseconds(delayMicros);
  digitalWrite(STEP_PIN, LOW);  digitalWrite(STEP2_PIN, LOW);
  delayMicroseconds(delayMicros);
```

Prepared by: Kenny Bui
Presented by: Kenny Bui

Engineering Analysis: Cleaning Activation

```
void enableMotors() {
  digitalWrite(ENABLE_PIN, LOW);
  digitalWrite(ENABLE2_PIN, LOW);
  Serial.println("Motors enabled.");
}

void stopMotors() {
  digitalWrite(ENABLE_PIN, HIGH);
  digitalWrite(ENABLE2_PIN, HIGH);
  Serial.println("Motors disabled.");
}

void startBrushMotor(bool forward) {
  if (forward) {
    digitalWrite(BRUSH_IN1, HIGH);
    digitalWrite(BRUSH_IN2, LOW);
  } else {
    digitalWrite(BRUSH_IN1, LOW);
    digitalWrite(BRUSH_IN2, HIGH);
  }
  analogWrite(BRUSH_EN, 255);
```

```
void stopBrushMotor() {
  digitalWrite(BRUSH_IN1, LOW);
  digitalWrite(BRUSH_IN2, LOW);
  analogWrite(BRUSH_EN, 0);
}

void buzzerAlert() {
  for (int i = 0; i < 10; i++) {
    digitalWrite(BUZZER, HIGH);
    delay(250);
    digitalWrite(BUZZER, LOW);
    delay(250);
```

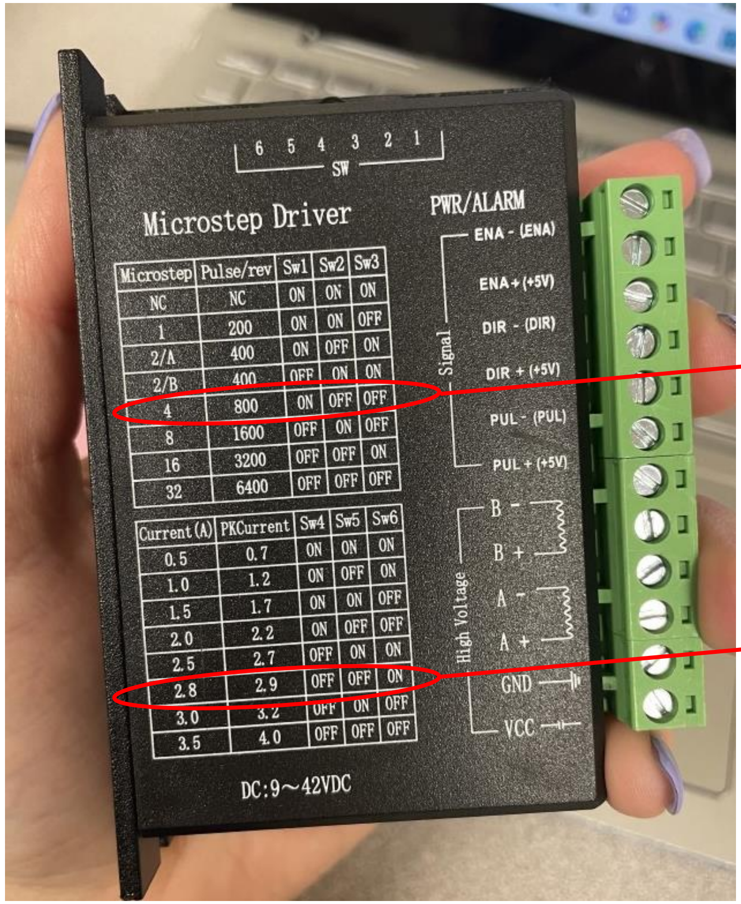
```
void emergencyStopAndWait() {
  stopBrushMotor();
  stopMotors();
  cancelTriggered = true;
  Serial.println("Cycle halted. Press START to return home.");

  while (digitalRead(START_BUTTON) == HIGH);
  while (digitalRead(START_BUTTON) == LOW);

  returnHome();
```

Prepared by: Margaret Burkes
Presented by: Margaret Burkes

TB6600 Dip Switch Settings

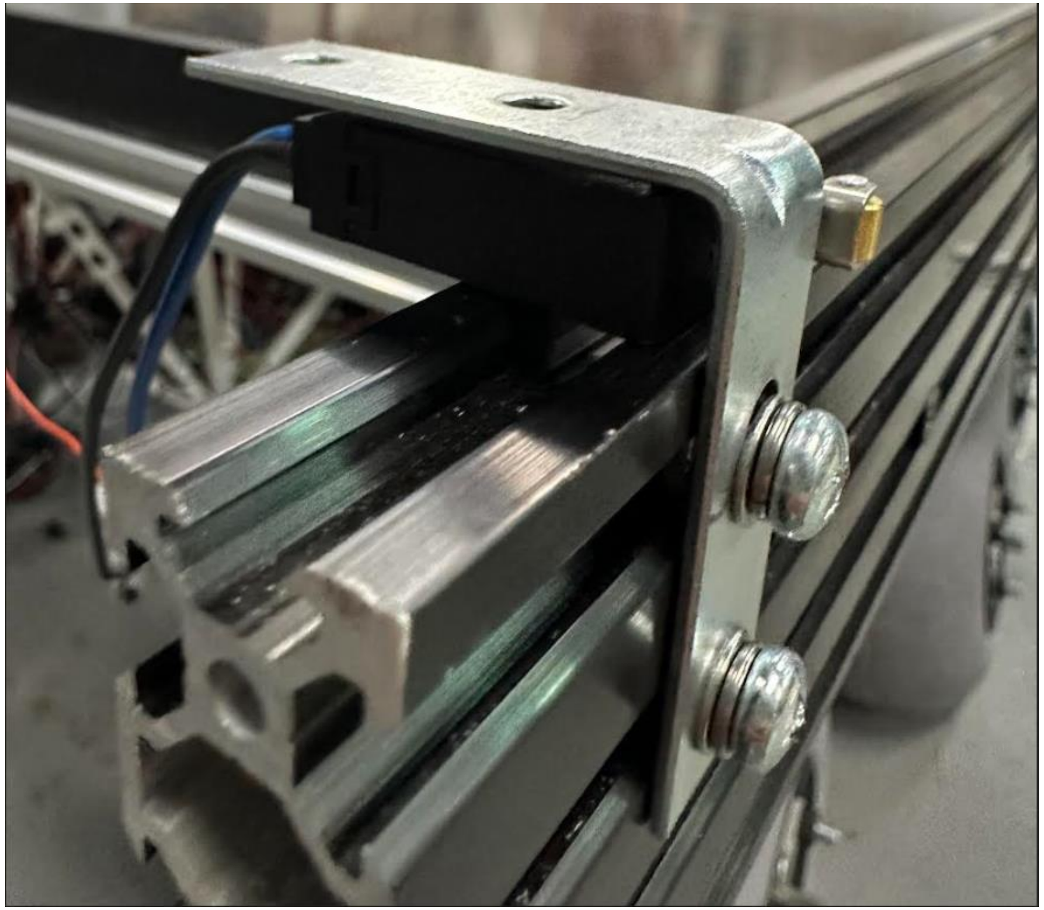


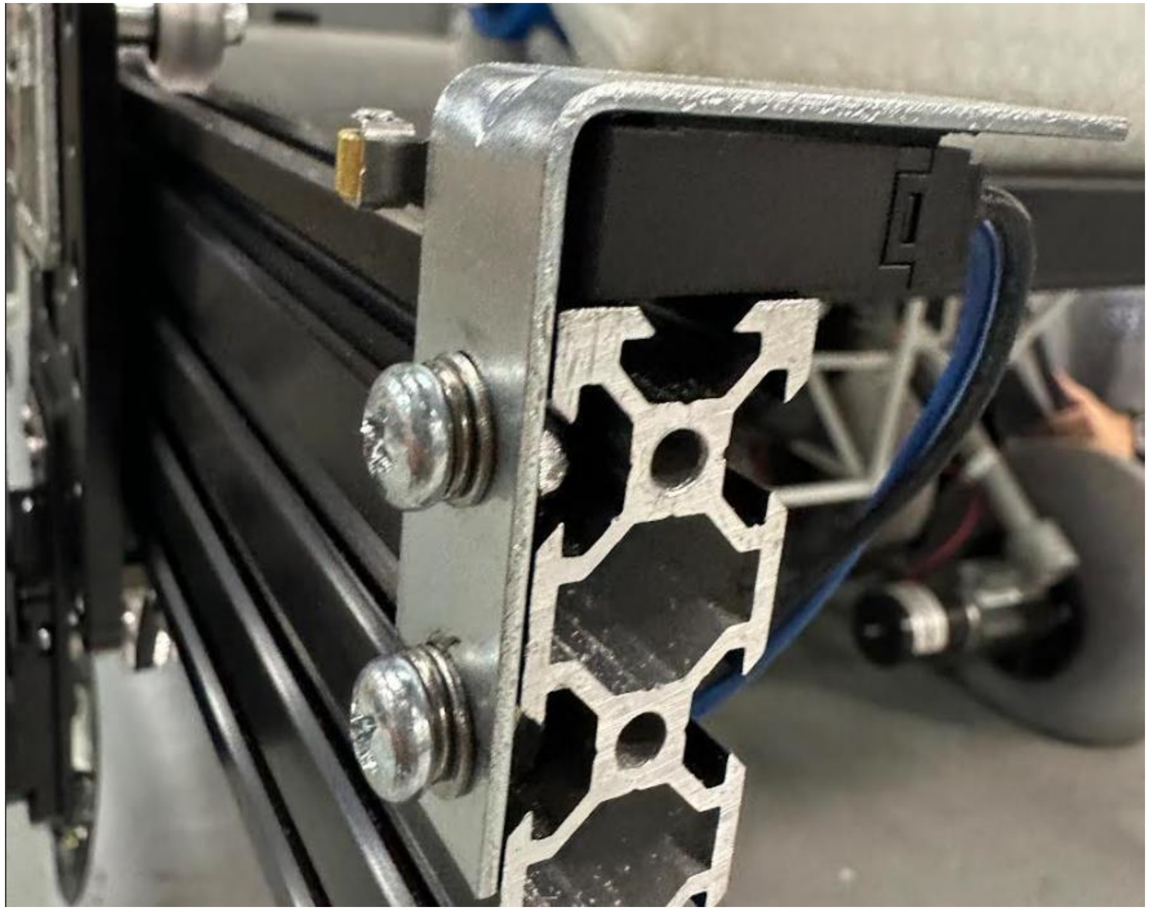
Quarter-Stepping Mode

Operating Current of the Motor is 2.8A

Prepared by: Kenny Bui
Presented by: Kenny Bui

Limit Switch Manufacturing

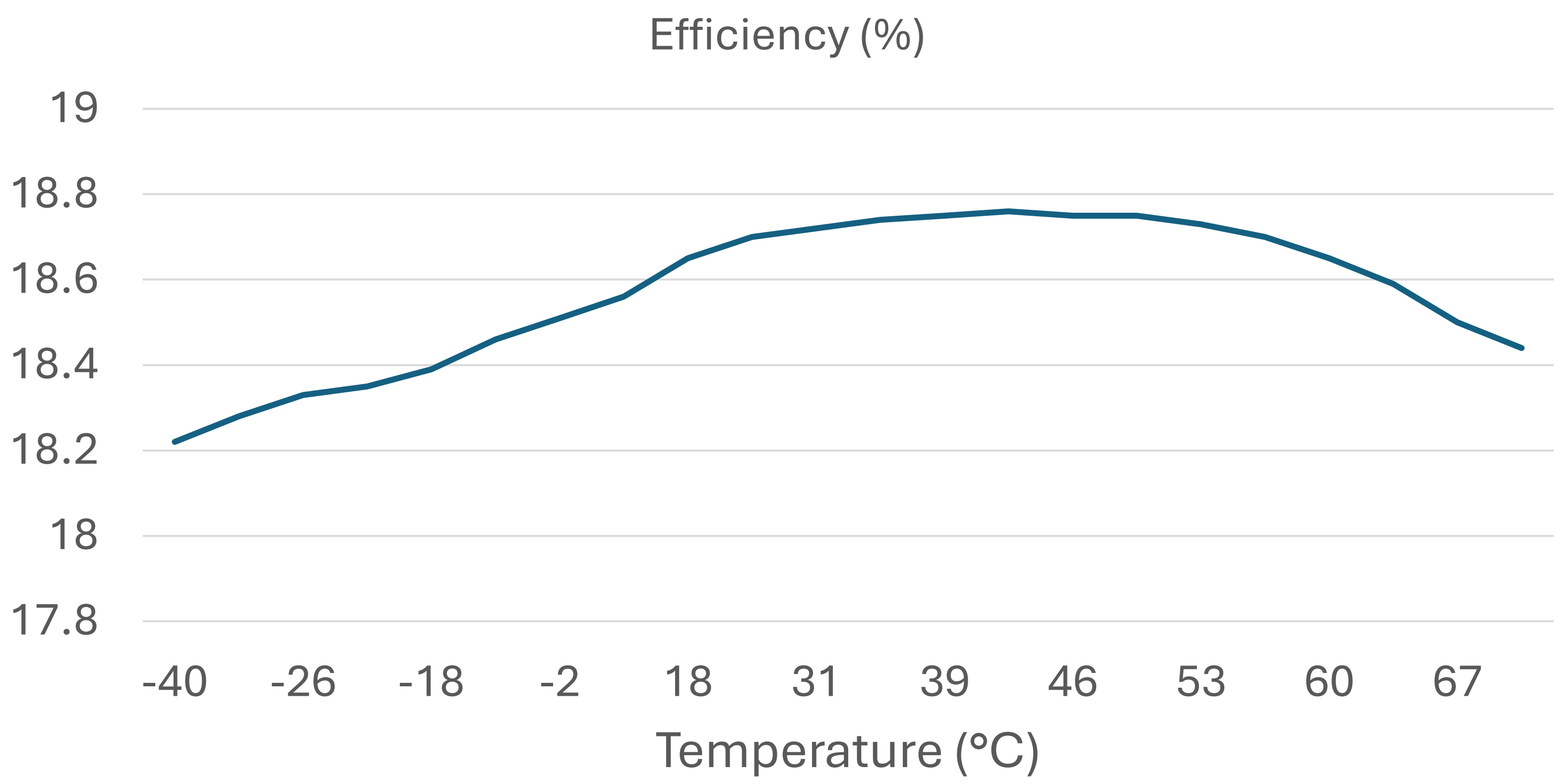




Prepared by: Bowen Williamson
Presented by: Bowen Williamson

Engineering Analysis: Charge Controller

Effects of Temperature on Solar Panel Efficiency



Prepared by: Kenny Bui
Presented by: Kenny Bui

Interior Space

Components	Volume (M^3)
L298N	0.000049851
TB6600 (2)	0.000354816
Arduino Uno	0.000054909
TriStar	0.005378
RD-1	0.0004346
Bus Bar (3)	0.0000648
	0.006336976

Original Volume	Occupied Space	
0.977	0.006336976	0.970663024

Prepared by: Keldon Ngo
Presented by: Keldon Ngo

Existing and Competing Technologies:

- Ecoppia E4:
  - Key similarities
    - Solar panel cleaning robot
    - Bi-directional cleaning
  - Key Differences
    - Smart row scanning
    - Speed adaptability

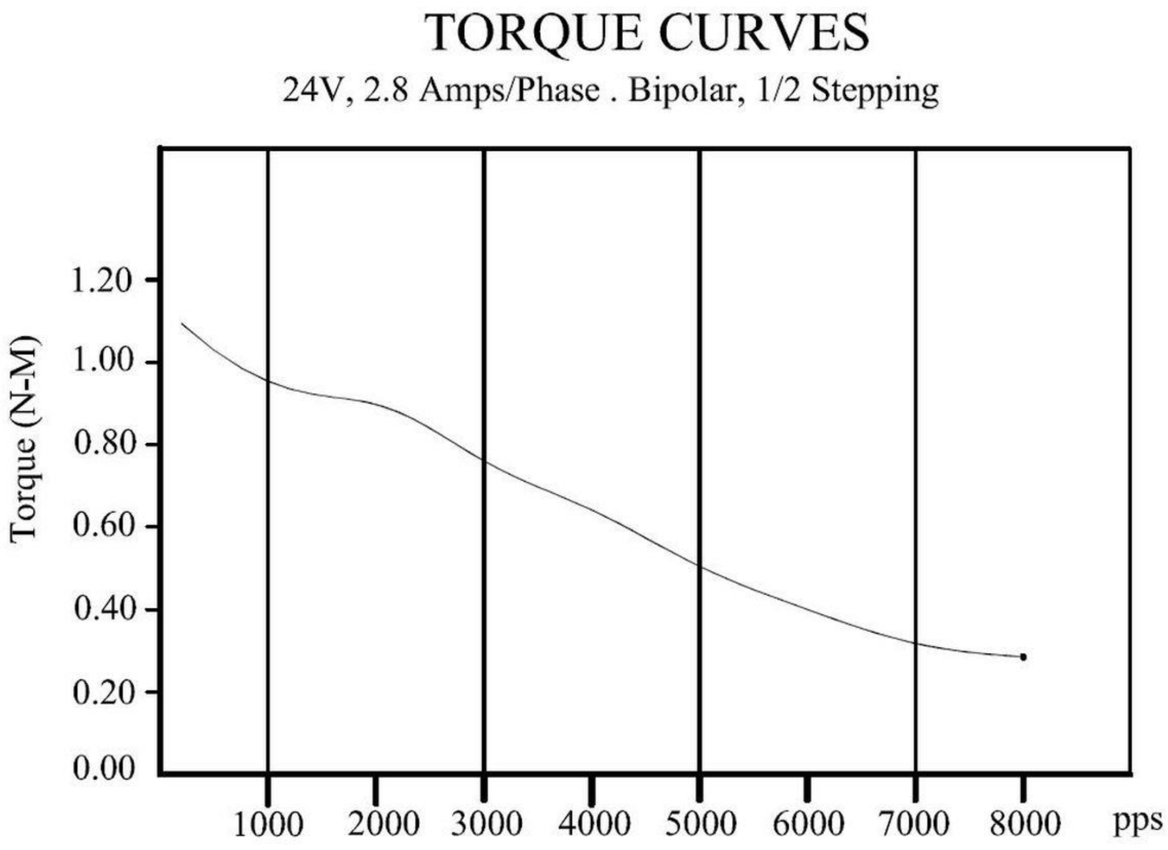
Prepared by: Keldon Ngo
Presented by: Keldon Ngo

Component Evaluation: Solar Panel

		Concepts		
Criterion	Weights	LG Neon	Candian Solar TOPHiKu6	REC Alpha
Size	40	0	0	1
Efficiency	40	-1	1	1
Cost	20	0	0	1
Weighted Total	100	-0.4	0.4	1

Prepared by: Margaret Burkes
Presented by: Margaret Burkes

Nema 23 Stepper Motor



Items	Specs
Phase Number	2 phases
Driving Voltage	12-48V
Resistance(20℃)	1.1$\pm$10% Ω / phase
Holding Torque	175 Oz.in
Detent Torque	400 g.cm
Rotor Inertia	300 g.cm^2
Step Angle	1.8° $\pm$ 5%
Rated Current	2.8A / phase
Inductance	3.0$\pm$20% mH/ phase
Weight	0.75 kg

Prepared by: William Mancuso
Presented by: William Mancuso

Concept Evaluation & Selection: Autonomous Power Switching

		Concepts		
Criterion	Weights	Voltage Regulator with Arduino	DC/DC Converter with Arduino	Solar Charge Controller
Functional Accuracy	45	-1	0	1
Simplicity	25	0	-1	1
Programmability	20	-1	0	1
Cost	10	1	0	-1
Weighted Total	100	-0.55	-0.25	0.8